EUROPEAN ORGANIZATION FOR NUCLEAR RESEARCH (CERN)

CERN-EP-20XX-ZZZ
LHCb-PAPER-2026-015
February 26, 2026

Angular analysis of the $B^+ \to \pi^+\mu^+\mu^-$ decay

LHCb collaboration[†]

Abstract

This paper presents a first measurement of the angular observables A_{FB} and F_{H} that parameterise the angular distribution of the $B^+ \to \pi^+\mu^+\mu^-$ decay. The pp collision dataset used in the analysis corresponds to an integrated luminosity of $9\,\mathrm{fb}^{-1}$, collected with the LHCb experiment between 2011 and 2018. The analysis is performed in two bins of dimuon invariant-mass squared, q^2, one above and one below the peaks in the q^2 distribution from the J/ψ and $\psi(2S)$ narrow charmonium resonances. The observables are found to be consistent with Standard Model predictions.

Submitted to JHEP

[†]Authors are listed at the end of this paper.

1 Introduction

The decay $B^+ \to \pi^+\mu^+\mu^-$ is, in the Standard Model of particle physics (SM), mediated by a b- to d-quark flavour-changing neutral-current transition.[1] Figure 1 shows the leading-order Feynman diagrams for the $B^+ \to \pi^+\mu^+\mu^-$ decay. The dominant contribution comes from diagrams with an internal t loop due to the large t mass, which enhances its virtual effects in the loop. The small magnitude of the V_{td} element of the Cabibbo–Kobayashi–Maskawa (CKM) quark-mixing matrix [1,2] suppresses the branching fraction for the decay to $\mathcal{O}(10^{-8})$. The small SM branching fraction makes the $B^+ \to \pi^+\mu^+\mu^-$ decay sensitive to contributions from particles arising in extensions of the SM. Non-SM contributions could alter observables such as the rate and angular distribution of the decay or their differences between B^+ and B^- decays. While previous measurements of the branching fraction and CP asymmetry [3, 4] are consistent with SM predictions [5–7], the angular distribution of the $B^+ \to \pi^+\mu^+\mu^-$ decay has not previously been measured.

The differential decay rate of the $B^+ \to \pi^+\mu^+\mu^-$ decay can be parameterised in terms of the dimuon mass squared, q^2, and the angle, θ_l, between the direction of the μ^+ and the direction opposite that of the pion in the dimuon rest frame for the B^+ meson decay. Normalising by the total decay rate Γ gives [8, 9]

$$\frac{1}{\Gamma}\frac{\mathrm{d}\Gamma}{\mathrm{d}\cos\theta_l} = \frac{3}{4}(1-F_H)(1-\cos^2\theta_l) + \frac{F_H}{2} + A_{\mathrm{FB}}\cos\theta_l\,. \tag{1}$$

This expression depends on two parameters, the forward-backward asymmetry of the dimuon system, A_{FB}, and a parameter that is a measure of the flatness of the $\cos\theta_l$ distribution, F_H. These parameters are sensitive to (pseudo)scalar and (axial)tensor contributions to the decay width [10]. In the SM, $A_{\mathrm{FB}} = 0$ and F_H is small away from both ends of the allowed q^2 range. For the probability density of the angular distribution to remain positive definite, $|A_{\mathrm{FB}}| \leq F_H/2$ is required. Small deviations from Eq. (1) due to quantum electrodynamics (QED) effects [11] are negligible in currently available data samples.

The angular distribution of the $B^+ \to \pi^+\mu^+\mu^-$ decay is studied for the first time in this paper using a dataset of proton-proton (pp) collisions corresponding to an integrated luminosity of $9\,\mathrm{fb}^{-1}$, collected with the LHCb experiment. The data are split into two regions of q^2, covering $1.1 < q^2 < 6.0\,\mathrm{GeV}^2/c^4$ and $15.0 < q^2 < 22.0\,\mathrm{GeV}^2/c^4$. While no

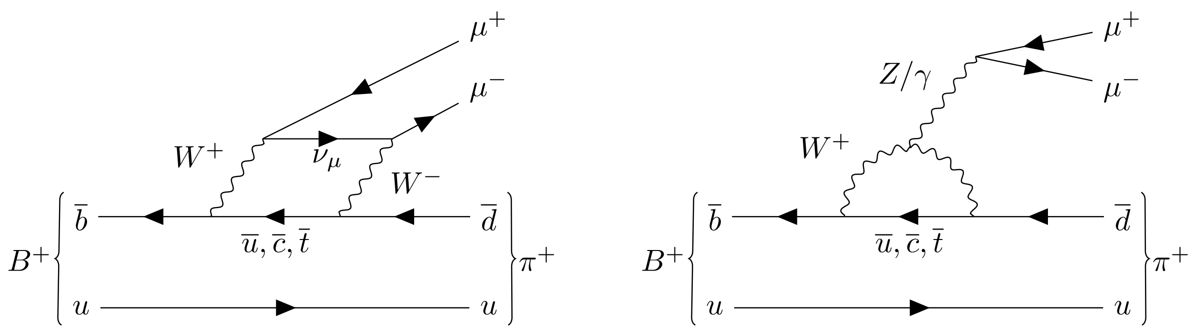

Figure 1: Leading-order Feynman diagrams for the $B^+ \to \pi^+\mu^+\mu^-$ decay in the Standard Model.

[1]The inclusion of charge-conjugate processes is implied throughout this paper unless explicitly stated.

measurements have previously been made of the angular distribution of the $B^+ \to \pi^+\mu^+\mu^-$ decay, the BaBar, Belle, CDF, CMS and LHCb collaborations have previously studied the corresponding distribution for $B^+ \to K^+\mu^+\mu^-$ decays [12–16]. This decay has a much larger branching fraction due to the larger magnitude of the V_{ts} element of the CKM matrix, relative to V_{td}, and in the SM is expected to have a similar angular distribution. The results of the $B^+ \to K^+\mu^+\mu^-$ analyses are consistent with the SM picture, with A_{FB} and F_H consistent with zero. These measurements have been used to set the best available limits on tensor contributions to $b \to s$ processes [10].

2 Detector and simulation

The LHCb detector [17, 18] is a single-arm forward spectrometer covering the pseudorapidity range $2 < \eta < 5$, designed for the study of particles containing b or c quarks. The detector to collect the data analysed in this paper includes a high-precision tracking system consisting of a silicon-strip vertex detector surrounding the pp interaction region [19], a large-area silicon-strip detector located upstream of a dipole magnet with a bending power of about 4 T m, and three stations of silicon-strip detectors and straw drift tubes [20, 21] placed downstream of the magnet. The tracking system provides a measurement of the momentum, p, of charged particles with a relative uncertainty that varies from 0.5% at low momentum to 1.0% at 200 GeV/c. The minimum distance of a track to a primary pp collision vertex (PV), the impact parameter (IP), is measured with a resolution of $(15 + 29/p_{\mathrm{T}})\,\mu\mathrm{m}$, where p_{T} is the component of the momentum transverse to the beam, in GeV/c. Different types of charged hadrons are distinguished using information from two ring-imaging Cherenkov detectors [22]. Photons, electrons and hadrons are identified by a calorimeter system consisting of scintillating-pad and preshower detectors, an electromagnetic and a hadronic calorimeter. Muons are identified by a system composed of alternating layers of iron and multiwire proportional chambers [23]. The online event selection is performed by a trigger [24], which consists of a hardware stage, based on information from the calorimeter and muon systems, followed by a software stage, which applies a full event reconstruction. Triggered data further undergo a centralised, offline processing step to deliver physics-analysis-ready data across the entire LHCb physics programme [25].

Samples of simulated events are used to understand the impact of the detector and subsequent event selection on the angular distribution of the signal $B^+ \to \pi^+\mu^+\mu^-$ decays. They are also used to model the contribution from a variety of backgrounds as described below. In the simulation, pp collisions are generated using Pythia [26] with a specific LHCb configuration [27]. Decays of unstable particles are described by EvtGen [28], in which final-state radiation is generated using Photos [29]. The angular and q^2 distributions of $B^+ \to \pi^+\mu^+\mu^-$ and $B^+ \to K^+\mu^+\mu^-$ decays are generated according to Ref. [9] using form factors from Ref. [30]. The interaction of the generated particles with the detector, and its response, are implemented using the Geant4 toolkit [31] as described in Ref. [32]. The simulated samples are corrected for known data-simulation differences in track reconstruction and muon-identification efficiencies using a tag-and-probe approach with $J/\psi \to \mu^+\mu^-$ decays. Kaon and pion identification efficiencies are corrected using $D^{*+} \to D^0(K^-\pi^+)\pi^+$ decays, which can be selected in data without particle-identification requirements. Simulated samples are then weighted such that the production kinematics of the B^+ mesons in the simulation are consistent with the observed momentum distributions of B^+ mesons reconstructed through $B^+ \to J/\psi K^+$decays in data.

3 Candidate selection

Events are required to pass a hardware trigger that retains events comprising at least one muon with high transverse momentum, with the minimum threshold varying in the range 1200–2200 MeV/c depending on the data-taking period. In the software trigger, at least one of the final-state particles is required to have $p_{\mathrm{T}} > 0.8\,\mathrm{GeV}/c$ and an IP greater than $100\,\mu\mathrm{m}$ with respect to all PVs in the event. A multivariate algorithm [33, 34] is then used to identify events containing secondary vertices formed by two or three tracks that are consistent with the decay of a b hadron.

Offline, B^+ candidates are formed from the combination of two oppositely charged tracks that are identified as muons and a third track identified as a pion. The muons are required to have $p_{\mathrm{T}} > 800\,\mathrm{MeV}/c$. Each track is required to have a good track-fit quality, to be within the acceptance of the particle-identification systems, and have a significant IP with respect to all PVs in the event. The three tracks are required to be consistent with originating from a single point that is displaced from every PV. The B^+ candidate is required to have a $\pi^+\mu^+\mu^-$ mass in the range $5180 < m(\pi^+\mu^+\mu^-) < 5780\,\mathrm{MeV}/c^2$ and to be consistent with originating from its associated PV, *i.e.* the PV with the smallest IP χ^2, defined as the difference in the vertex-fit χ^2 of the PV reconstructed with and without the candidate.

Combinatorial background, from tracks originating from two or more different b- or c-hadron decays, is suppressed using a boosted-decision-tree classifier. The classifier is trained using samples of simulated decays for signal and candidates from sidebands of the data with $\pi^+\mu^+\mu^-$ mass in the range $5550 < m(\pi^+\mu^+\mu^-) < 6000\,\mathrm{MeV}/c^2$ or $6500 < m(\pi^+\mu^+\mu^-) < 7000\,\mathrm{MeV}/c^2$ for background. The range $6000 < m(\pi^+\mu^+\mu^-) < 6500\,\mathrm{MeV}/c^2$ is excluded to avoid potential contributions from resonant or nonresonant $B_c^+ \to \pi^+\mu^+\mu^-$ decays [35]. The classifier takes as input a variety of topological and kinematic variables that are characteristic of B^+ meson decays in the LHCb experiment. These include the muon and pion momenta, the muon IP χ^2, the B^+ candidate vertex-fit χ^2 and the angle between the B^+ momentum-vector and the vector between the B^+ decay vertex and its associated PV. The classifier working point and particle-identification requirements applied to the pion are optimised simultaneously using pseudoexperiments to maximise the expected significance of the $B^+ \to \pi^+\mu^+\mu^-$ signal.

The largest potential source of background from a specific decay arises from the $B^+ \to K^+\mu^+\mu^-$ channel, where the kaon is incorrectly identified as a pion. Before applying particle-identification requirements, this background is 25 times more prominent than the signal [36]. Residual background contributions arising from $B^+ \to \pi^+\pi^-\pi^+$ and $B^+ \to K^+\pi^-\pi^+$ decays amount to approximately 1% of the signal yield; these contributions are neglected in the nominal analysis but are accounted for as a source of systematic uncertainty. Backgrounds are also present from $B^+ \to J/\psi K^+$ and $B^+ \to J/\psi\pi^+$ decays, where the kaon or pion is incorrectly identified as a muon and a muon from the $J/\psi \to \mu^+\mu^-$ decay, with the appropriate charge, is misidentified as a pion. These are removed by vetoing candidates with a $\pi^+\mu^-$ mass, with the muon mass hypothesis assigned to the pion, within $20\,\mathrm{MeV}/c^2$ of the known J/ψ mass [36]; this is subsequently referred to as the J/ψ veto. Decays of B^0, B^+ and B_s^0 mesons to final states containing a π^+, μ^+, μ^- and additional unreconstructed particles predominantly populate the $\pi^+\mu^+\mu^-$ invariant-mass region below $5180\,\mathrm{MeV}/c^2$. These contributions are removed by the requirement on the $\pi^+\mu^+\mu^-$ mass and are therefore not considered further. To suppress further sources of semileptonic backgrounds with missing neutrinos, candidates are vetoed if they have a $\pi^+\mu^-$ mass that is consistent with a D meson. The largest residual contribution from decays with unreconstructed particles within the considered $\pi^+\mu^+\mu^-$ mass range is from $B_s^0 \to f_0(980)\mu^+\mu^-$ decays. Given its small branching fraction [37], this background can be safely neglected in the analysis.

Samples of $B^+ \to J/\psi\pi^+$ and $B^+ \to J/\psi K^+$decays, where the J/ψ meson decays to $\mu^+\mu^-$, are retained as control channels. The $B^+ \to J/\psi\pi^+$ decay is selected with the same set of requirements as the $B^+ \to \pi^+\mu^+\mu^-$ signal, except requiring the $\mu^+\mu^-$ mass to be within $100\,\mathrm{MeV}/c^2$ of the known J/ψ mass instead of the q^2 requirements. The $B^+ \to J/\psi K^+$decay is selected with an alternative set of particle-identification requirements.

4 Angular model

In each q^2 interval, the observables A_{FB} and F_H are determined by performing a simultaneous unbinned maximum-likelihood fit to the $\pi^+\mu^+\mu^-$ mass and $\cos\theta_l$ distributions of the selected candidates. The signal angular distribution is modelled by Eq. (1), multiplied by a function describing the angular efficiency of the reconstruction and selection process. The angular resolution of the detector is sufficiently good (better than 1 mrad) that it can be neglected but is considered as a source of systematic uncertainty.

Figure 2 shows the efficiency of the reconstruction and selection, without the vetoes applied, as a function of $\cos\theta_l$ in the two q^2 ranges on simulated $B^+ \to \pi^+\mu^+\mu^-$ decays. The efficiency is well described in each q^2 interval by a fourth-order Legendre polynomial model that only includes even-ordered terms and is therefore symmetric around $\cos\theta_l = 0$. The variation of $B^+ \to \pi^+\mu^+\mu^-$ kinematics over q^2 causes the selection efficiency of the rare mode to differ between low and high q^2 intervals. The main sources of inefficiency are related to the muons needing to have sufficient momentum to reach and traverse the muon system and to satisfy the hardware trigger requirements. At low q^2, the track and muon reconstruction requirements favour decays with θ_l away from 0 and π, where one muon is soft in the B^+ rest frame. At intermediate and high q^2, the hardware trigger requirements on the p_{T} of the muon candidates favour events away from $\theta_l \sim \pi/2$, where the maximum muon momentum tends to be smaller.

The simulation is generated with an SM-like q^2 distribution [30], and potential variations of the efficiency profile within each q^2 interval are considered as a source of systematic uncertainty. The J/ψ veto removes, at a fixed q^2 value, a narrow region of $\cos\theta_l$, which cannot be modelled by a polynomial. The efficiency is multiplied by a discontinuous function to account for this effect.

The signal mass distribution is described by the sum of a Gaussian function with power-law tails on both left- and right-hand sides of the distribution and a second Gaussian function. The tail parameters are fixed from fits to simulation. In the fits to data, an offset in the position of the peak and a scale factor on the width of the distribution, due

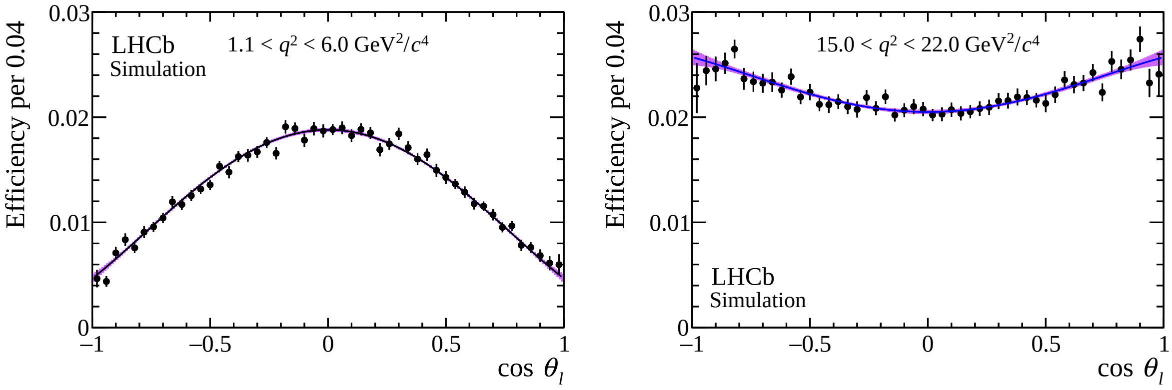

Figure 2: Efficiency as a function of $\cos\theta_l$ for $B^+ \to \pi^+\mu^+\mu^-$ decays in simulation in the q^2 regions (left) $1.1 < q^2 < 6.0\,\mathrm{GeV}^2/c^4$ and (right) $15.0 < q^2 < 22.0\,\mathrm{GeV}^2/c^4$. The projections of fits using a fourth-order polynomial parameterisation, containing only even-ordered terms, are superimposed, with the uncertainty on the obtained efficiency function shown as a band.

to momentum-scale uncertainties, are included and determined from fits to the $K^+\mu^+\mu^-$ mass distribution of selected $B^+ \to J/\psi K^+$decays.

The background from misidentified $B^+ \to K^+\mu^+\mu^-$ decays is modelled using simulation generated according to the SM [9,30], as the distribution in data is known to be consistent with SM expectations [38]. The angular distribution is described by a fourth-order Chebyshev polynomial function, with no contribution from odd-ordered terms. The mass distribution of the misidentified background is described by a Gaussian function with power-law tails on both left- and right-hand sides of the distribution. The mass and angular distributions are assumed to factorise in the fit, and a systematic uncertainty is assigned due to this assumption.

Combinatorial background is described by an exponential function in $\pi^+\mu^+\mu^-$ mass and a second-order Chebyshev polynomial function in $\cos\theta_l$. The parameters of these distributions are free to vary in the fit to data.

Due to the small size of the samples in the two q^2 intervals, and due to the physical boundary $|A_{\mathrm{FB}}| \leq F_H/2$, the Feldman–Cousins approach [39] is used to estimate confidence intervals on A_{FB} and F_H. At each fixed value of F_H and A_{FB}, the best-fit value for the nuisance parameters is used when generating pseudoexperiments. When presenting intervals on A_{FB} (F_H), F_H (A_{FB}) is treated as a nuisance parameter.

A fit is performed to the selected $B^+ \to J/\psi\pi^+$ control samples in order to validate the analysis strategy. The model used to describe $B^+ \to J/\psi\pi^+$ decays is identical to that of the signal but with the vetoes removed and the efficiency parameterised over the appropriate q^2 range. The $B^+ \to J/\psi\pi^+$ decay has a known angular distribution, proportional to $\sin^2\theta_l$. This is a signal-like distribution but with $A_{\mathrm{FB}} = 0$ and $F_H = 0$. A fit is also performed to the sample of $B^+ \to J/\psi K^+$decays. In this fit only contributions from signal and combinatorial background are considered.

5 Systematic uncertainties

The impact of various sources of systematic uncertainty are estimated using pseudoexperiments generated assuming datasets that are three orders of magnitude larger than the

Table 1: Sources of systematic uncertainty on F_H and A_{FB} in the $1.1 < q^2 < 6.0\,\mathrm{GeV}^2/c^4$ ($15.0 < q^2 < 22.0\,\mathrm{GeV}^2/c^4$) regions. The different contributions are described in the text, and the total uncertainty is obtained adding the individual sources of uncertainty in quadrature.

Source	F_H	A_{FB}
Simulation q^2 dependence	0.12 (0.01)	0.01 (0.01)
Angular efficiency order	0.01 (0.01)	0.01 (0.01)
Simulation sample size	0.01 (0.01)	0.01 (0.02)
Data-simulation corrections	0.01 (0.02)	0.03 (0.04)
Truth matching criteria	0.01 (0.01)	0.01 (0.02)
Angular resolution	0.01 (0.01)	0.01 (0.02)
$B^+ \to K^+\mu^+\mu^-$ background model	0.01 (0.01)	0.01 (0.02)
Combinatorial background model	0.01 (0.01)	0.01 (0.02)
Hadronic backgrounds	0.04 (0.02)	0.01 (0.01)
$B^0_s \to f_0(980)\mu^+\mu^-$ background	0.01 (0.02)	0.01 (0.02)
Total	0.14 (0.04)	0.04 (0.07)

data. In most cases, the experiments are generated according to an alternative hypothesis and then fit using the nominal analysis strategy and a systematic uncertainty is assigned based on the bias on the fitted observables added together with the uncertainty on the observable in quadrature. Exceptions to this process are discussed below. The different sources of systematic uncertainty are summarised in Table 1.

Five types of systematic uncertainty are considered related to the signal efficiency model. The uncertainty associated with the knowledge of the q^2 spectrum of the signal is the largest source of systematic uncertainty on F_H. To assign this uncertainty, the simulation is weighted to match the q^2 spectrum observed in data within each q^2 interval and a pseudoexperiment is generated according to the angular efficiency of the weighted simulation. The uncertainty accounting for the fixed efficiency model order is assigned by considering alternative models: one that allows odd-ordered terms and another that allows for an additional term of order six. The uncertainty due to the limited size of the simulated samples used to derive the efficiency model is evaluated by bootstrapping the simulation and deriving a set of varied models. Pseudoexperiments are then generated from each efficiency model and fit with the nominal model. The spread in values of the observables from this set of pseudoexperiments is assigned as a systematic uncertainty. A similar approach is used to derive an uncertainty associated with the data-simulation corrections. Potential imperfections in the matching of true decays to reconstructed candidates in the simulation is a further source of uncertainty. To assess this uncertainty, a pseudoexperiment is generated according to an efficiency model derived with a more relaxed set of matching criteria on the pion and muon tracks.

Five sources of systematic uncertainty are considered related to choices made in modelling the mass and angular distributions of the signal and background components. To assess the impact of neglecting the signal angular resolution, a pseudoexperiment is generated with the θ_l distribution broadened according to the resolution obtained in the simulation. The systematic uncertainty associated to the assumption that the $\pi^+\mu^+\mu^-$ mass and $\cos\theta_l$ distributions factorise for misidentified $B^+ \to K^+\mu^+\mu^-$ decays is assessed by producing a pseudoexperiment in which the $B^+ \to K^+\mu^+\mu^-$ background component is modelled nonparametrically with events sampled from simulation. The uncertainty due to the combinatorial background model is assessed by relaxing the classifier requirement and fitting the upper mass sideband of the data using a fourth-order polynomial function. The shape parameters from this fit are subsequently used when generating the combinatorial background component of a pseudoexperiment. Finally, uncertainties due to neglecting hadronic backgrounds from $B^+ \to K^+\pi^-\pi^+$, $B^+ \to \pi^+\pi^-\pi^+$and $B^0_s \to f_0(980)\mu^+\mu^-$ decays are determined by including additional background components in the pseudoexperiment generation. The mass and angular distribution of the hadronic background components are based on simulated $B^+ \to K^+\pi^-\pi^+$ and $B^+ \to \pi^+\pi^-\pi^+$decays that are weighted to reproduce the Dalitz structure of the decays measured by BaBar [40] and LHCb [41], respectively. The yield of the hadronic background is determined using control regions of the data with relaxed muon-identification requirements.

6 Results

The $\pi^+\mu^+\mu^-$ mass and $\cos\theta_l$ distributions of selected $B^+ \to J/\psi\pi^+$ candidates are shown in Fig. 3. Excellent agreement is found between the data and the result of the fit, indicating that the signal efficiency is well described by the corrected simulation. The F_H and $A_{\rm FB}$ parameters are found to be consistent with zero, as expected for these decays. Similarly, F_H and $A_{\rm FB}$ determined from the fit to the $B^+ \to J/\psi K^+$sample are found to be consistent with zero.

The $\pi^+\mu^+\mu^-$ mass and $\cos\theta_l$ distributions of the selected candidates in the two q^2 regions are shown in Fig. 4. The data are described well by the result of the fit. The discontinuity in $\cos\theta_l$ (visible in the $1.1 < q^2 < 6.0\,{\rm GeV}^2/c^4$ range) arises from the J/ψ veto. The observed signal yields are 89 ± 13 and 135 ± 15 events in the low- and high-q^2 regions, respectively. The uncertainties quoted on the yields are purely statistical and are determined by repeating the fit on pseudodata generated by bootstrapping the data set [42]. Figure 5 shows the one-dimensional $\cos\theta_l$ distributions within the mass interval $5260 < m(\pi^+\mu^+\mu^-) < 5300\,{\rm MeV}/c^2$, corresponding to a window of approximately twice the $\pi^+\mu^+\mu^-$ mass resolution around the known B^+ meson mass.

The values of $A_{\rm FB}$ and F_H for the two q^2 intervals are given in Table 2 and are shown in Fig. 6. Two-dimensional confidence interval regions for $A_{\rm FB}$ and F_H are shown in Fig. 7. At low q^2, the data show a non-zero $A_{\rm FB}$ value and a large value of F_H that are not consistent with the SM point within 95% confidence level (CL), but are consistent within 99% CL. At high q^2, the data show values of $A_{\rm FB}$ and F_H that are consistent with the SM point within 68% CL.

Stability tests of the results are conducted by imposing alternative selection criteria on the output of the boosted-decision-tree background classifier, by applying more stringent particle-identification requirements, and by determining F_H independently of $A_{\rm FB}$ through an alternative fit to the distribution of the absolute value of $\cos\theta_l$. All these studies yield results that are consistent with the baseline measurements.

As an additional test, the $\cos\theta_l$ distribution in data was examined for $B^+ \to K^+\mu^+\mu^-$ decays reconstructed in the low- and high-q^2 regions. The candidates were selected using the same criteria as for the signal mode, with the exception of the particle-identification requirements on the charged kaon, for which the criteria used in the selection

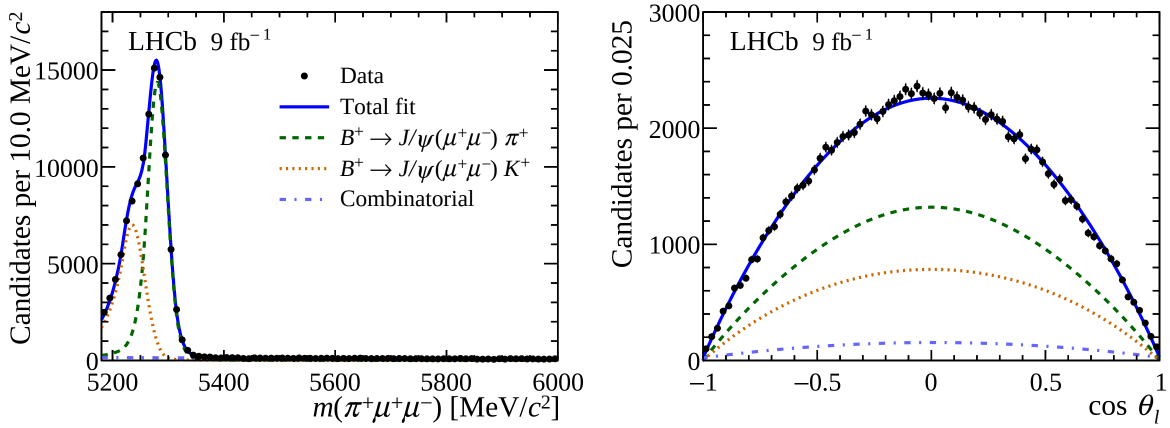

Figure 3: Distribution of (left) the $\pi^+\mu^+\mu^-$ mass and (right) $\cos\theta_l$ for selected $B^+\to J/\psi\pi^+$ candidates. The data are overlaid with the result of the fit described in the text.

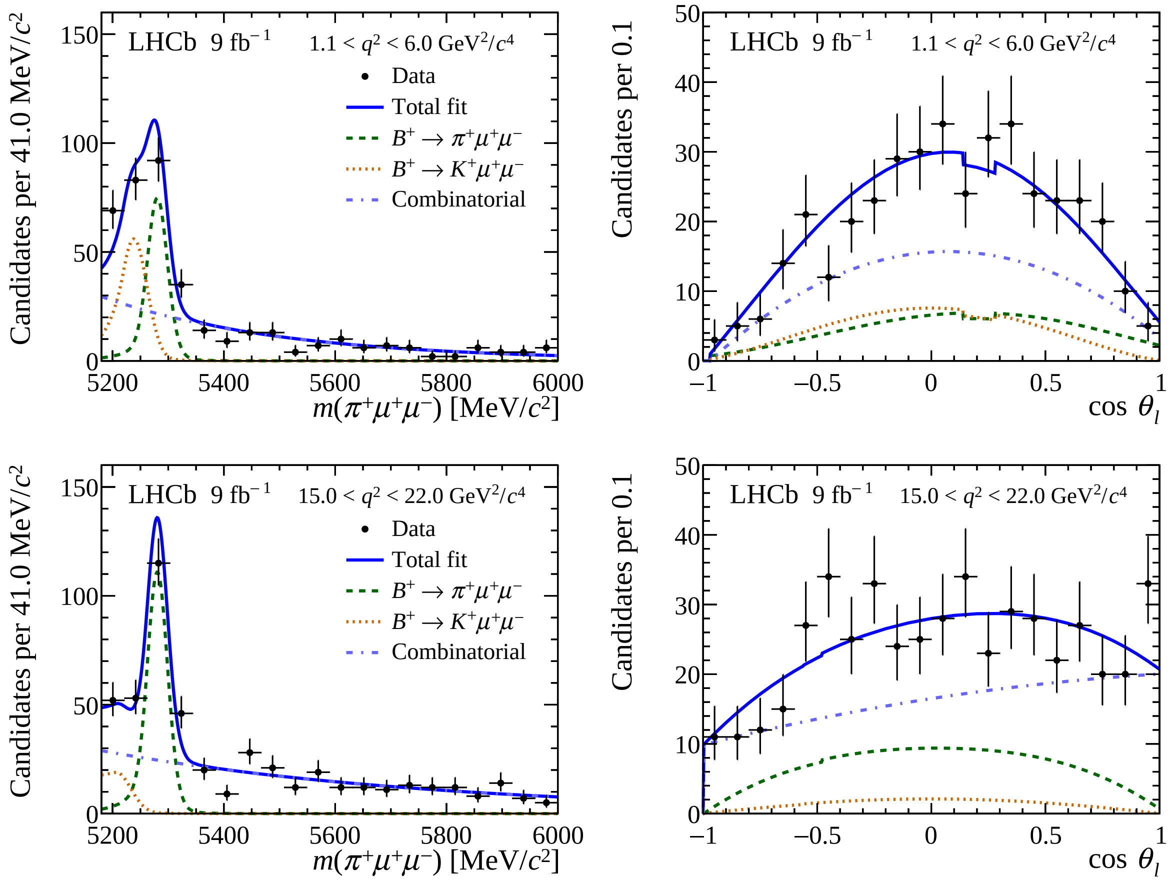

Figure 4: Distribution of (left) the $\pi^+\mu^+\mu^-$ mass and (right) $\cos\theta_l$ for selected $B^+\to\pi^+\mu^+\mu^-$ candidates in the range (top) $1.1 < q^2 < 6.0$ and (bottom) $15.0 < q^2 < 22.0\,\mathrm{GeV}^2/c^4$. The data are overlaid with the result of the fit described in the text.

of $B^+\to J/\psi K^+$decays were applied. In both the low- and high-q^2 regions, no significant asymmetry was observed for $B^+\to K^+\mu^+\mu^-$ decays.

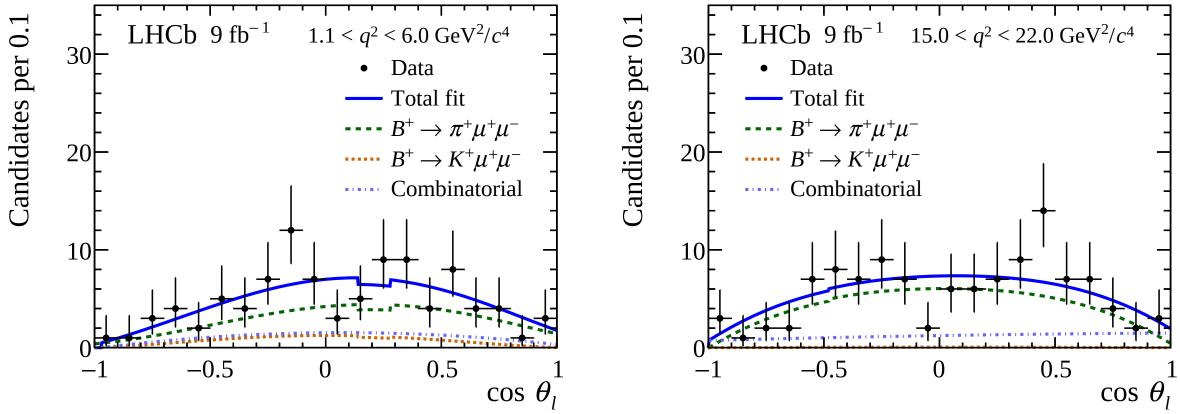

Figure 5: Distribution of $\cos\theta_l$ for selected $B^+ \to \pi^+\mu^+\mu^-$ candidates in the range (left) $1.1 < q^2 < 6.0$ and (right) $15.0 < q^2 < 22.0\,\text{GeV}^2/c^4$. The candidates are required to be within the mass interval $5260 < m(\pi^+\mu^+\mu^-) < 5300\,\text{MeV}/c^2$, corresponding to a window of approximately twice the $\pi^+\mu^+\mu^-$ mass resolution around the known B^+ meson mass. The data are overlaid with the result of the fit described in the text.

Table 2: Values of A_{FB} and F_H obtained from the fit and Feldman–Cousins procedure. The intervals correspond to 68% confidence level.

	F_H	F_H interval	A_{FB}	A_{FB} interval
$1.1 < q^2 < 6.0\,\text{GeV}^2/c^4$	0.91	$[0.60, 0.99]$	0.27	$[\ \ 0.11, 0.42]$
$15.0 < q^2 < 22.0\,\text{GeV}^2/c^4$	0.04	$[0.00, 0.20]$	0.02	$[-0.02, 0.09]$

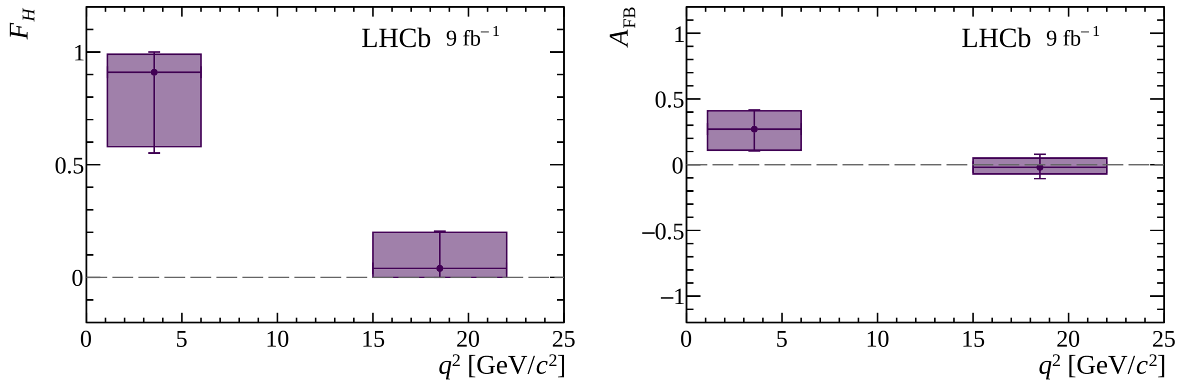

Figure 6: Parameters F_H and A_{FB} in bins of dimuon mass squared, q^2. The marker indicates the best-fit value from the likelihood fit. The boxes represent the 68% confidence level intervals as determined using the Feldman–Cousins construction. The outer bars represent the total uncertainty, obtained by adding in quadrature the additional contribution arising from the systematic uncertainty of the measurement.

7 Summary

The angular distribution of the $B^+ \to \pi^+\mu^+\mu^-$ decay is studied for the first time using a dataset of pp collisions corresponding to $9\,\text{fb}^{-1}$ of integrated luminosity, collected with the

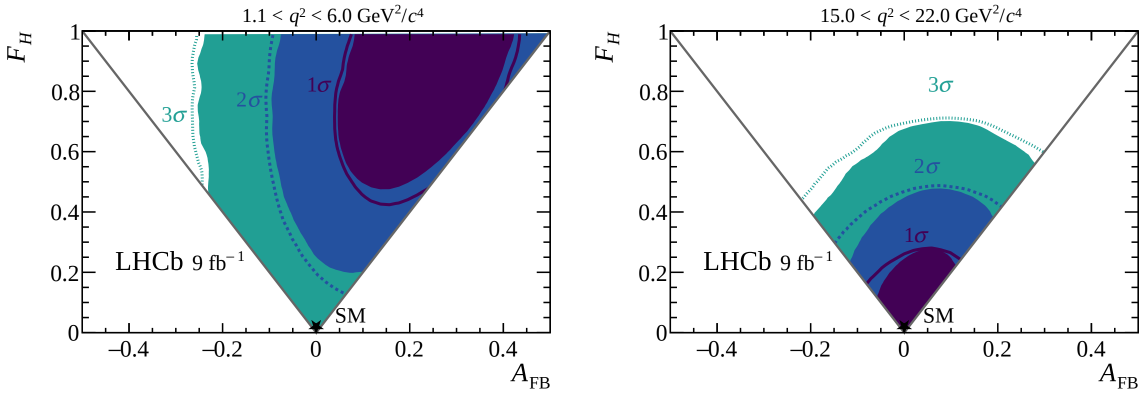

Figure 7: Two-dimensional confidence level regions obtained with the Feldman–Cousins procedure for A_{FB} and F_H in the q^2 regions (left) $1.1 < q^2 < 6.0$ and (right) $15.0 < q^2 < 22.0\,\mathrm{GeV}^2/c^4$. The one-, two-, and three-sigma regions are indicated in violet, blue, and green, respectively, corresponding to the associated statistical intervals. The line contours represent the corresponding confidence levels obtained when systematic uncertainties are combined in quadrature. The boundary of the physical region in which the probability distribution of the signal remains positive for all values of $\cos\theta_l$ is shown by the gray lines.

LHCb experiment. The angular observables A_{FB} and F_H, which parameterise the angular distribution of the decay, are found to be consistent with Standard Model (SM) predictions at the 99% confidence level (CL) in the dimuon mass squared region $1.1 < q^2 < 6.0\,\mathrm{GeV}^2/c^4$ and at the 68% CL in the region $15.0 < q^2 < 22.0\,\mathrm{GeV}^2/c^4$. These findings are in agreement with previous determinations of the corresponding observables in the $B^+ \to K^+\mu^+\mu^-$ decay [16].

Acknowledgements

We express our gratitude to our colleagues in the CERN accelerator departments for the excellent performance of the LHC. We thank the technical and administrative staff at the LHCb institutes. We acknowledge support from CERN and from the national agencies: ARC (Australia); CAPES, CNPq, FAPERJ and FINEP (Brazil); MOST and NSFC (China); CNRS/IN2P3 (France); BMBF, DFG and MPG (Germany); INFN (Italy); NWO (Netherlands); MNiSW and NCN (Poland); MCID/IFA (Romania); MICIU and AEI (Spain); SNSF and SER (Switzerland); NASU (Ukraine); STFC (United Kingdom); DOE NP and NSF (USA). We acknowledge the computing resources that are provided by ARDC (Australia), CBPF (Brazil), CERN, IHEP and LZU (China), IN2P3 (France), KIT and DESY (Germany), INFN (Italy), SURF (Netherlands), Polish WLCG (Poland), IFIN-HH (Romania), PIC (Spain), CSCS (Switzerland), and GridPP (United Kingdom). We are indebted to the communities behind the multiple open-source software packages on which we depend. Individual groups or members have received support from Key Research Program of Frontier Sciences of CAS, CAS PIFI, CAS CCEPP, Fundamental Research Funds for the Central Universities, and Sci. & Tech. Program of Guangzhou (China); Minciencias (Colombia); EPLANET, Marie Skłodowska-Curie Actions, ERC and NextGenerationEU (European Union); A*MIDEX, ANR, IPhU and Labex P2IO, and Région Auvergne-Rhône-Alpes (France); Alexander-von-Humboldt Foundation (Germany);

ICSC (Italy); Severo Ochoa and María de Maeztu Units of Excellence, GVA, XuntaGal, GENCAT, InTalent-Inditex and Prog. Atracción Talento CM (Spain); SRC (Sweden); the Leverhulme Trust, the Royal Society and UKRI (United Kingdom).

Supplementary material for LHCb-PAPER-2026-015

Figure 8 shows the signal selection efficiency, after applying the complete set of selection criteria, as a two-dimensional function of q^2 and $\cos\theta_l$. Figure 9 shows the $K^+\mu^+\mu^-$ mass and $\cos\theta_l$ distributions of selected $B^+\to J/\psi K^+$decays. The observables $A_{\rm FB}$ and F_H are determined by performing a fit to the mass and $\cos\theta_l$ distributions. The signal model is modified to remove the J/ψ veto and the angular efficiency is derived from simulated $B^+\to J/\psi K^+$decays over the appropriate q^2 range. Only the combinatorial-background component is included. Background from incorrectly identified $B^+\to J/\psi\pi^+$ decays, which populates the region to the right of the $B^+\to J/\psi K^+$, signal is small and is neglected. The resulting values of $A_{\rm FB}$ and F_H are consistent with zero, as expected. The difference in negative log-likelihood of the fit between the best-fit point and the tested point in $A_{\rm FB}$ and F_H is shown for selected $B^+\to J/\psi K^+$and $B^+\to J/\psi\pi^+$ decays in Fig. 10.

Table 3 lists the fitted values of the parameters obtained from fits to the $B^+\to\pi^+\mu^+\mu^-$ data in both the low- and high-q^2 regions. Figure 11 presents the confidence intervals for F_H and $A_{\rm FB}$ obtained using the Feldman–Cousins construction, where the parameters are treated independently in the interval determination. Figure 12 shows the raw forward-backward asymmetry in the number of forward and backward going μ^+,

$$a_{\rm FB} = \frac{N(\cos\theta_l > 0) - N(\cos\theta_l < 0)}{N(\cos\theta_l > 0) + N(\cos\theta_l < 0)}\,,$$

as a function of the $\pi^+\mu^+\mu^-$ mass.

Figure 13 shows the fit projections of a two-dimensional fit to the joint distribution of the $\pi^+\mu^+\mu^-$ mass and the absolute value of $\cos\theta_l$, performed as a consistency check to extract F_H using an alternative estimation strategy that does not depend on $A_{\rm FB}$. Figure 14 shows the one-dimensional distribution of the absolute value of $\cos\theta_l$ within the mass interval $5260 < m(\pi^+\mu^+\mu^-) < 5300\,{\rm MeV}/c^2$. In this case, the angular distribution normalised by the total decay rate, is given by

$$\frac{1}{\Gamma}\frac{{\rm d}\Gamma}{{\rm d}|\cos\theta_l|} = \frac{3}{2}(1-F_{\rm H})(1-|\cos\theta_l|^2) + F_{\rm H}\,.$$

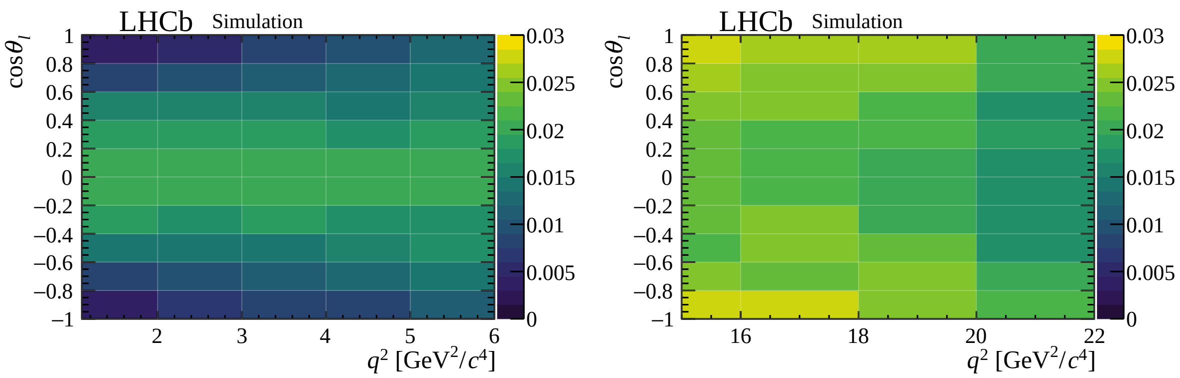

Figure 8: Efficiency as a function of q^2 and $\cos\theta_l$ for $B^+\to\pi^+\mu^+\mu^-$ decays in simulation in the q^2 regions (left) $1.1 < q^2 < 6.0$ and (right) $15.0 < q^2 < 22.0\,{\rm GeV}^2/c^4$.

Table 3: Results of the fit to the $B^+ \to \pi^+\mu^+\mu^-$ data set. The fit results are reported for the signal angular coefficients $A_{\rm FB}$ and F_H, for the event yields of the signal component and of the misidentified $B^+ \to K^+\mu^+\mu^-$ and combinatorial background components, for the slope parameter (e_0) of the exponential model describing the combinatorial $m(\pi^+\mu^+\mu^-)$ distribution, and for the parameters (p_1 and p_2) associated with the first and second coefficients of a second-order Chebyshev polynomial used to model the combinatorial angular distribution. The quoted uncertainties are purely statistical and are obtained from the likelihood minimisation procedure. For the high-q^2 interval, the covariance matrix is not positive definite and therefore the uncertainties should be interpreted with caution.

Parameter	$1.1 < q^2 < 6.0\,\mathrm{GeV}^2/c^4$	$15.0 < q^2 < 22.0\,\mathrm{GeV}^2/c^4$
$A_{\rm FB}$	0.26 ± 0.15	0.02 ± 0.20
F_H	0.91 ± 0.50	0.04 ± 0.81
Signal $B^+ \to \pi^+\mu^+\mu^-$ yield	90 ± 15	134 ± 30
Misidentified $B^+ \to K^+\mu^+\mu^-$ yield	86 ± 17	27 ± 22
Combinatorial background yield	216 ± 22	320 ± 24
Combinatorial e_0 $[(\mathrm{GeV}/c^2)^{-1}]$	-3.06 ± 0.50	-1.62 ± 1.03
Combinatorial p_1	0.25 ± 0.16	0.32 ± 0.15
Combinatorial p_2	-0.86 ± 0.13	-0.05 ± 0.29

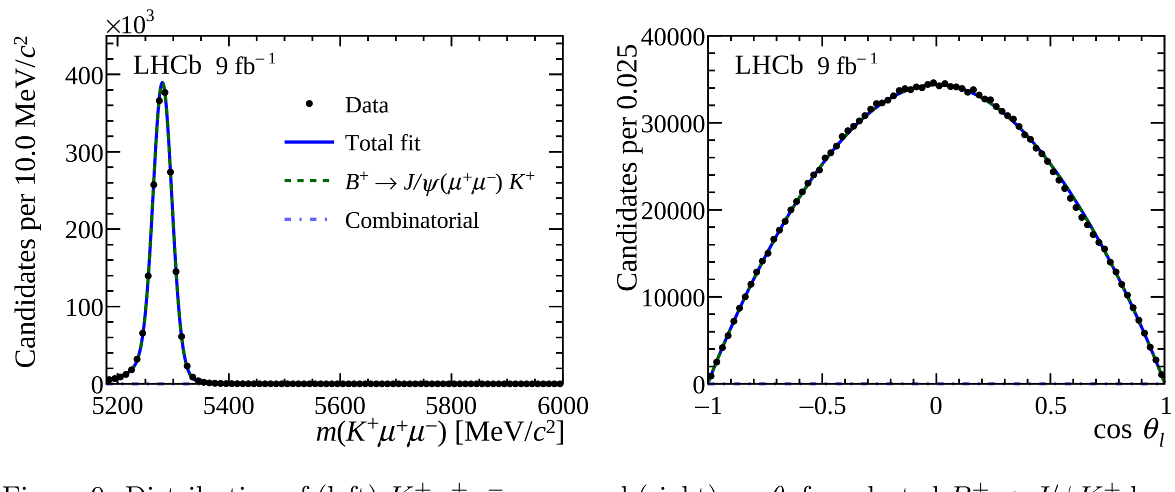

Figure 9: Distribution of (left) $K^+\mu^+\mu^-$ mass and (right) $\cos\theta_l$ for selected $B^+ \to J/\psi K^+$decays. The data are overlaid with the result of the fit described in the text. The contribution from combinatorial background is negligible and thus barely visible.

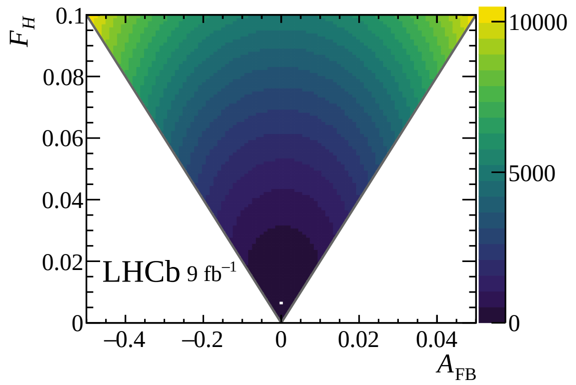

Figure 10: Difference in negative log-likelihood between the best fit and the tested point as a function of A_{FB} and F_H from the fits to selected (left) $B^+ \to J/\psi K^+$and (right) $B^+ \to J/\psi \pi^+$ decays.

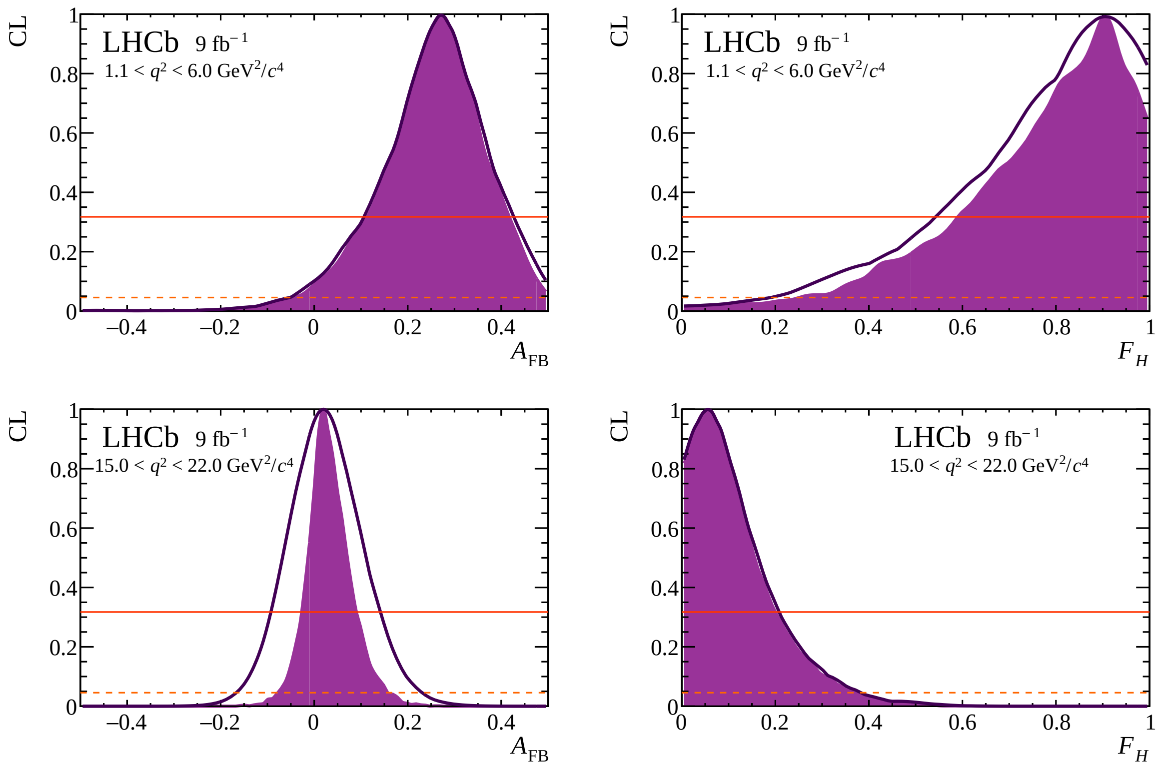

Figure 11: Confidence level from the Feldman-Cousins procedure for (left) A_{FB} and (right) F_H for q^2 in the range (top) $1.1 < q^2 < 6.0$ and (bottom) $15.0 < q^2 < 22.0\,\mathrm{GeV}^2/c^4$. The confidence level for each observable is obtained by treating the other one as a nuisance parameter. The filled violet region denotes the confidence intervals derived using the Feldman–Cousins construction and reflects only statistical uncertainties. The solid violet curve indicates the corresponding confidence intervals obtained when statistical and systematic uncertainties are combined in quadrature. The horizontal solid red and dashed orange lines indicate the one- and two-standard-deviation confidence levels, respectively.

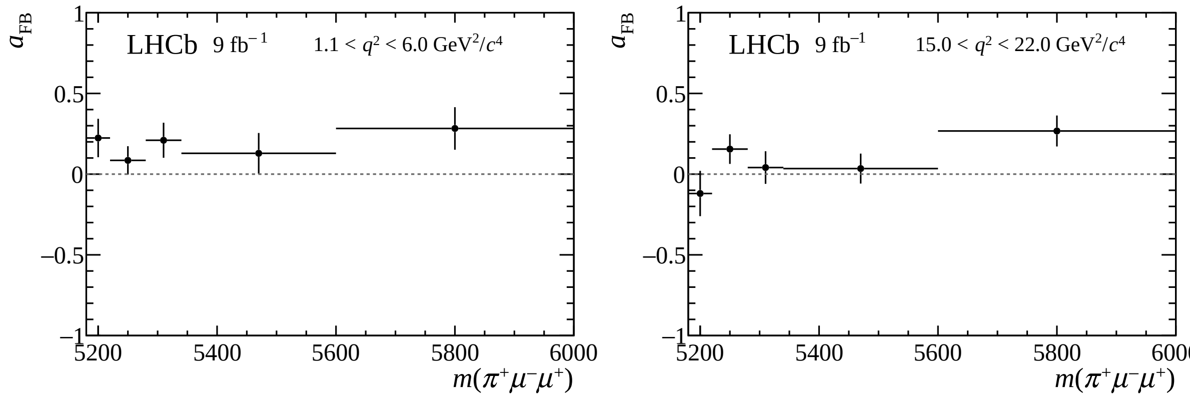

Figure 12: Raw forward-backward asymmetry for selected $B^+ \to \pi^+\mu^+\mu^-$ candidates in the range (left) $1.1 < q^2 < 6.0$ and (right) $15.0 < q^2 < 22.0\,\text{GeV}^2/c^4$.

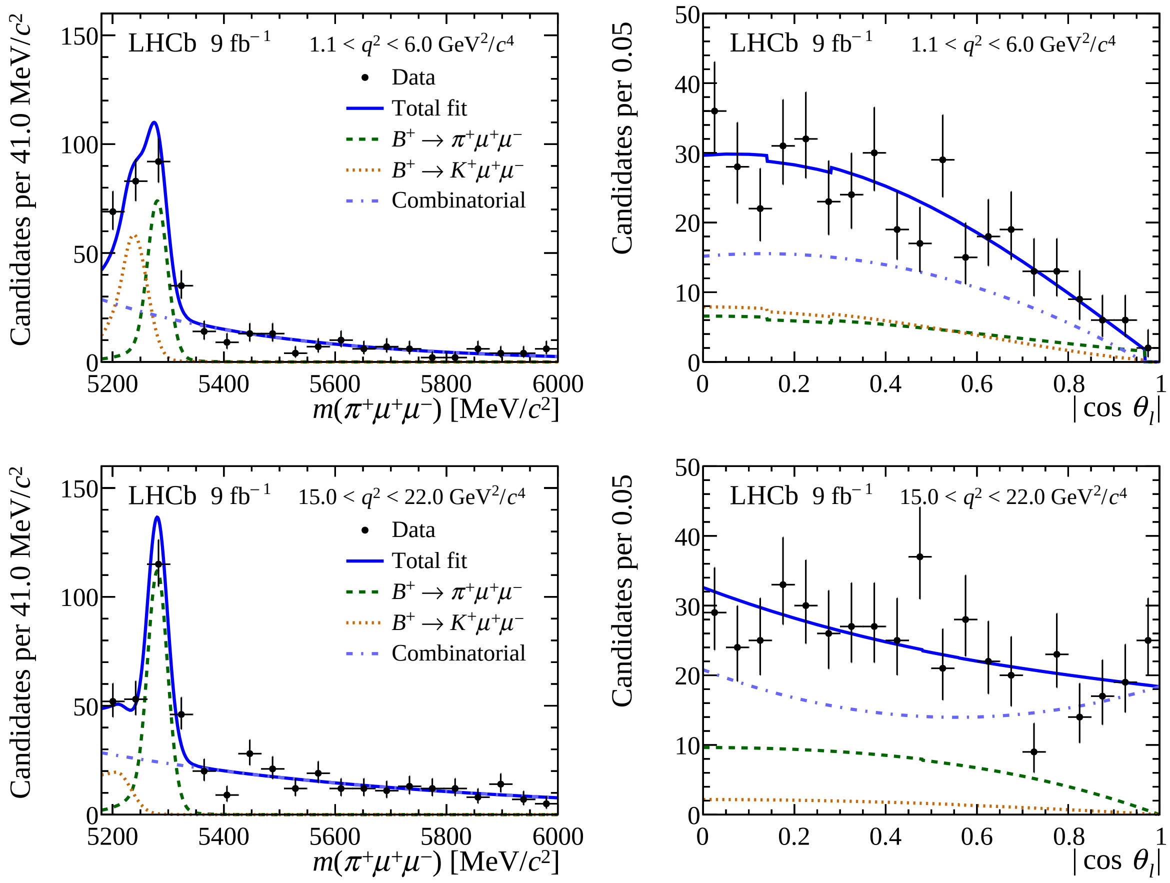

Figure 13: Distribution of (left) the $\pi^+\mu^+\mu^-$ mass and (right) absolute value of $\cos\theta_l$ for selected $B^+ \to \pi^+\mu^+\mu^-$ candidates in the range (top) $1.1 < q^2 < 6.0$ and (bottom) $15.0 < q^2 < 22.0\,\text{GeV}^2/c^4$. The data are overlaid with the result of the fit described in the text. The discontinuity in $\cos\theta_l$ (visible in the $1.1 < q^2 < 6.0\,\text{GeV}^2/c^4$ range) arises from the J/ψ veto.

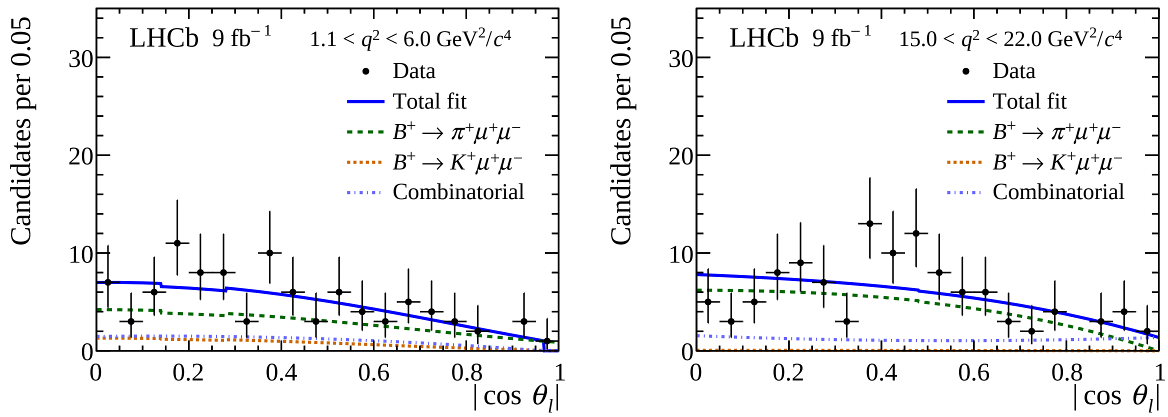

Figure 14: Distribution of the absolute value of $\cos\theta_l$ for selected $B^+ \to \pi^+\mu^+\mu^-$ candidates in the range (left) $1.1 < q^2 < 6.0$ and (right) $15.0 < q^2 < 22.0\,\mathrm{GeV}^2/c^4$. The candidates are required to be within the mass interval $5260 < m(\pi^+\mu^+\mu^-) < 5300\,\mathrm{MeV}/c^2$, corresponding to a window of approximately twice the $\pi^+\mu^+\mu^-$ mass resolution, centred on the known B^+ mass. The data are overlaid with the result of the fit described in the text. The discontinuity in $\cos\theta_l$ (visible in the $1.1 < q^2 < 6.0\,\mathrm{GeV}^2/c^4$ range) arises from the J/ψ veto.

References

[1] N. Cabibbo, *Unitary symmetry and leptonic decays*, Phys. Rev. Lett. **10** (1963) 531.

[2] M. Kobayashi and T. Maskawa, *CP-violation in the renormalizable theory of weak interaction*, Prog. Theor. Phys. **49** (1973) 652.

[3] LHCb collaboration, R. Aaij *et al.*, *First observation of the decay $B^+ \to \pi^+\mu^+\mu^-$*, JHEP **12** (2012) 125, `arXiv:1210.2645`.

[4] LHCb collaboration, R. Aaij *et al.*, *First measurement of the differential branching fraction and CP asymmetry of the $B^+ \to \pi^+\mu^+\mu^-$ decay*, JHEP **10** (2015) 034, `arXiv:1509.00414`.

[5] A. Ali, A. Y. Parkhomenko, and A. V. Rusov, *Precise calculation of the dilepton invariant-mass spectrum and the decay rate in $B^\pm \to \pi^\pm\mu^+\mu^-$ in the SM*, Phys. Rev. **D89** (2014) 094021, `arXiv:1312.2523`.

[6] C. Hambrock, A. Khodjamirian, and A. Rusov, *Hadronic effects and observables in $B \to \pi\ell^+\ell^-$ decay at large recoil*, Phys. Rev. **D92** (2015) 074020, `arXiv:1506.07760`.

[7] Fermilab Lattice and MILC collaborations, J. A. Bailey *et al.*, *$B \to \pi\ell\ell$ form factors for new-physics searches from lattice QCD*, Phys. Rev. Lett. **115** (2015) 152002, `arXiv:1507.01618`.

[8] C. Bobeth, G. Hiller, and G. Piranishvili, *Angular distributions of $\bar{B} \to \bar{K}\ell^+\ell^-$ decays*, JHEP **12** (2007) 040, `arXiv:0709.4174`.

[9] A. Ali, P. Ball, L. T. Handoko, and G. Hiller, *A comparative study of the decays $B \to (K, K^*)\ell^+\ell^-$ in standard model and supersymmetric theories*, Phys. Rev. **D61** (2000) 074024, `arXiv:hep-ph/9910221`.

[10] F. Beaujean, C. Bobeth, and S. Jahn, *Constraints on tensor and scalar couplings from $B \to K\bar{\mu}\mu$ and $B_s \to \bar{\mu}\mu$*, Eur. Phys. J. **C75** (2015) 456, `arXiv:1508.01526`.

[11] G. Isidori, S. Nabeebaccus, and R. Zwicky, *QED corrections in $\overline{B} \to \overline{K}\ell^+\ell^-$ at the double-differential level*, JHEP **12** (2020) 104, `arXiv:2009.00929`.

[12] BaBar collaboration, B. Aubert *et al.*, *Measurements of branching fractions, rate asymmetries, and angular distributions in the rare decays $B \to K\ell^+\ell^-$ and $B \to K^*\ell^+\ell^-$*, Phys. Rev. **D73** (2006) 092001, `arXiv:hep-ex/0604007`.

[13] Belle collaboration, J.-T. Wei *et al.*, *Measurement of the differential branching fraction and forward-backward asymmetry for $B \to K^{(*)}\ell^+\ell^-$*, Phys. Rev. Lett. **103** (2009) 171801, `arXiv:0904.0770`.

[14] CDF collaboration, T. Aaltonen *et al.*, *Measurements of the angular distributions in the decays $B \to K^{(*)}\mu^+\mu^-$ at CDF*, Phys. Rev. Lett. **108** (2012) 081807, `arXiv:1108.0695`.

[15] CMS collaboration, A. M. Sirunyan *et al.*, *Angular analysis of the decay $B^+ \to K^+\mu^+\mu^-$ in proton-proton collisions at $\sqrt{s} = 8$ TeV*, Phys. Rev. **D98** (2018) 112011, arXiv:1806.00636.

[16] LHCb collaboration, R. Aaij *et al.*, *Angular analysis of charged and neutral $B \to K\mu^+\mu^-$ decays*, JHEP **05** (2014) 082, arXiv:1403.8045.

[17] LHCb collaboration, A. A. Alves Jr. *et al.*, *The LHCb detector at the LHC*, JINST **3** (2008) S08005.

[18] LHCb collaboration, R. Aaij *et al.*, *LHCb detector performance*, Int. J. Mod. Phys. **A30** (2015) 1530022, arXiv:1412.6352.

[19] R. Aaij *et al.*, *Performance of the LHCb Vertex Locator*, JINST **9** (2014) P09007, arXiv:1405.7808.

[20] R. Arink *et al.*, *Performance of the LHCb Outer Tracker*, JINST **9** (2014) P01002, arXiv:1311.3893.

[21] P. d'Argent *et al.*, *Improved performance of the LHCb Outer Tracker in LHC Run 2*, JINST **12** (2017) P11016, arXiv:1708.00819.

[22] M. Adinolfi *et al.*, *Performance of the LHCb RICH detector at the LHC*, Eur. Phys. J. **C73** (2013) 2431, arXiv:1211.6759.

[23] A. A. Alves Jr. *et al.*, *Performance of the LHCb muon system*, JINST **8** (2013) P02022, arXiv:1211.1346.

[24] R. Aaij *et al.*, *The LHCb trigger and its performance in 2011*, JINST **8** (2013) P04022, arXiv:1211.3055.

[25] N. Grieser *et al.*, *The LHCb stripping project: Sustainable legacy data processing for high-energy physics*, Comput. Softw. Big. Sci. **9** (2025) 21, arXiv:2509.05294.

[26] T. Sjöstrand, S. Mrenna, and P. Skands, *A brief introduction to PYTHIA 8.1*, Comput. Phys. Commun. **178** (2008) 852, arXiv:0710.3820.

[27] I. Belyaev *et al.*, *Handling of the generation of primary events in Gauss, the LHCb simulation framework*, J. Phys. Conf. Ser. **331** (2011) 032047.

[28] D. J. Lange, *The EvtGen particle decay simulation package*, Nucl. Instrum. Meth. **A462** (2001) 152.

[29] N. Davidson, T. Przedzinski, and Z. Was, *PHOTOS interface in C++: Technical and physics documentation*, Comp. Phys. Comm. **199** (2016) 86, arXiv:1011.0937.

[30] P. Ball and R. Zwicky, *New results on $B \to \pi, K, \eta$ decay form factors from light-cone sum rules*, Phys. Rev. **D71** (2005) 014015, arXiv:hep-ph/0406232.

[31] Geant4 collaboration, J. Allison *et al.*, *Geant4 developments and applications*, IEEE Trans. Nucl. Sci. **53** (2006) 270; Geant4 collaboration, S. Agostinelli *et al.*, *Geant4: A simulation toolkit*, Nucl. Instrum. Meth. **A506** (2003) 250.

[32] M. Clemencic *et al.*, *The LHCb simulation application, Gauss: Design, evolution and experience*, J. Phys. Conf. Ser. **331** (2011) 032023.

[33] V. V. Gligorov and M. Williams, *Efficient, reliable and fast high-level triggering using a bonsai boosted decision tree*, JINST **8** (2013) P02013, arXiv:1210.6861.

[34] T. Likhomanenko *et al.*, *LHCb topological trigger reoptimization*, J. Phys. Conf. Ser. **664** (2015) 082025, arXiv:1510.00572.

[35] LHCb collaboration, R. Aaij *et al.*, *Search for $B^{*0}_{(s)} \to \mu^+\mu^-$ in $B_c^+ \to \pi^+\mu^+\mu^-$ decays*, Eur. Phys. J **C85** (2025) 20, arXiv:2409.17209.

[36] Particle Data Group, S. Navas *et al.*, *Review of particle physics*, Phys. Rev. **D110** (2024) 030001.

[37] LHCb collaboration, R. Aaij *et al.*, *Study of the rare B^0_s and B^0 decays into the $\pi^+\pi^-\mu^+\mu^-$ final state*, Phys. Lett. **B743** (2015) 46, arXiv:1412.6433.

[38] LHCb collaboration, R. Aaij *et al.*, *Study of the kinematic dependences of Λ^0_b production in pp collisions and a measurement of the $\Lambda^0_b \to \Lambda^+_c\pi^-$ branching fraction*, JHEP **08** (2014) 143, arXiv:1405.6842.

[39] G. J. Feldman and R. D. Cousins, *A Unified approach to the classical statistical analysis of small signals*, Phys. Rev. **D57** (1998) 3873, arXiv:physics/9711021.

[40] BaBar collaboration, B. Aubert *et al.*, *Evidence for direct CP violation from Dalitz-plot analysis of $B^\pm \to K^\pm\pi^\mp\pi^\pm$*, Phys. Rev. **D78** (2008) 012004, arXiv:0803.4451.

[41] LHCb collaboration, R. Aaij *et al.*, *Amplitude analysis of the $B^+ \to \pi^+\pi^+\pi^-$ decay*, Phys. Rev. **D101** (2020) 012006, arXiv:1909.05212.

[42] B. Efron, *Bootstrap methods: Another look at the jackknife*, Ann. Statist. **7** (1979) 1.

LHCb collaboration

R. Aaij[38], M. Abdelfatah[69], A.S.W. Abdelmotteleb[57], C. Abellan Beteta[51], F. Abudinén[59], T. Ackernley[61], A. A. Adefisoye[69], B. Adeva[47], M. Adinolfi[55], P. Adlarson[87,42], C. Agapopoulou[14], C.A. Aidala[89], S. Akar[11], K. Akiba[38], P. Albicocco[28], J. Albrecht[19,f], R. Aleksiejunas[81], F. Alessio[49], P. Alvarez Cartelle[47], S. Amato[3], J.L. Amey[55], Y. Amhis[14], L. An[6], L. Anderlini[27], M. Andersson[51], P. Andreola[51], M. Andreotti[26], S. Andres Estrada[44], A. Anelli[31,o], D. Ao[7], C. Arata[12], F. Archilli[37], Z. Areg[69], M. Argenton[26], S. Arguedas Cuendis[9,49], L. Arnone[31,o], M. Artuso[69], E. Aslanides[13], R. Ataíde Da Silva[50], M. Atzeni[65], B. Audurier[12], J. A. Authier[15], D. Bacher[64], I. Bachiller Perea[50], S. Bachmann[22], M. Bachmayer[50], J.J. Back[57], Z. B. Bai[8], V. Balagura[15], A. Balboni[26], W. Baldini[26], Z. Baldwin[79], L. Balzani[19], H. Bao[7], J. Baptista de Souza Leite[2], C. Barbero Pretel[47,12], M. Barbetti[27], I. R. Barbosa[70], R.J. Barlow[63,†], M. Barnyakov[25], S. Baron[49], S. Barsuk[14], W. Barter[59], J. Bartz[69], S. Bashir[40], B. Batsukh[82], P. B. Battista[14], A. Bavarchee[80], A. Bay[50], A. Beck[65], M. Becker[19], F. Bedeschi[35], I.B. Bediaga[2], N. A. Behling[19], S. Belin[47], A. Bellavista[25], I. Belov[29], I. Belyaev[36], G. Bencivenni[28], E. Ben-Haim[16], J.L.M. Berkey[68], R. Bernet[51], A. Bertolin[33], F. Betti[59], J. Bex[56], O. Bezshyyko[88], S. Bhattacharya[80], M.S. Bieker[18], N.V. Biesuz[26], A. Biolchini[38], M. Birch[62], F.C.R. Bishop[10], A. Bitadze[63], A. Bizzeti[27,p], T. Blake[57,b], F. Blanc[50], J.E. Blank[19], S. Blusk[69], J.A. Boelhauve[19], O. Boente Garcia[49], T. Boettcher[90], A. Bohare[59], C. Bolognani[19], R. Bolzonella[26,l], R. B. Bonacci[1], A. Bordelius[49], F. Borgato[33,49], S. Borghi[63], M. Borsato[31,o], J.T. Borsuk[86], E. Bottalico[61], S.A. Bouchiba[50], M. Bovill[64], T.J.V. Bowcock[61], A. Boyer[49], C. Bozzi[26], J. D. Brandenburg[91], A. Brea Rodriguez[50], N. Breer[19], C. Breitfeld[19], J. Brodzicka[41], J. Brown[61], D. Brundu[32], E. Buchanan[59], M. Burgos Marcos[84], C. Burr[49], C. Buti[27], J.S. Butter[56], J. Buytaert[49], W. Byczynski[49], S. Cadeddu[32], H. Cai[75], Y. Cai[5], A. Caillet[16], R. Calabrese[26,l], L. Calefice[45], M. Calvi[31,o], M. Calvo Gomez[46], P. Camargo Magalhaes[2,a], J. I. Cambon Bouzas[47], P. Campana[28], A. C. Campos[3], A.F. Campoverde Quezada[7], Y. Cao[6], S. Capelli[31,o], M. Caporale[25], L. Capriotti[33], R. Caravaca-Mora[9], A. Carbone[25,j], L. Carcedo Salgado[47], R. Cardinale[29,m], A. Cardini[32], P. Carniti[31], L. Carus[22], A. Casais Vidal[65], R. Caspary[22], G. Casse[61], M. Cattaneo[49], G. Cavallero[26], V. Cavallini[26,l], S. Celani[49], I. Celestino[35,s], S. Cesare[49,n], A.J. Chadwick[61], I. Chahrour[89], M. Charles[16], Ph. Charpentier[49], E. Chatzianagnostou[38], R. Cheaib[80], M. Chefdeville[10], C. Chen[57], J. Chen[50], S. Chen[5], Z. Chen[7], A. Chen Hu[62], M. Cherif[12], S. Chernyshenko[53], X. Chiotopoulos[84], G. Chizhik[1], V. Chobanova[44], M. Chrzaszcz[41], V. Chulikov[28,36,49], P. Ciambrone[28], M.F. Cicala[57], X. Cid Vidal[47], P. Cifra[49], P.E.L. Clarke[59], M. Clemencic[49], H.V. Cliff[56], J. Closier[49], C. Cocha Toapaxi[22], V. Coco[49], J. Cogan[13], E. Cogneras[11], L. Cojocariu[43], S. Collaviti[50], P. Collins[49], T. Colombo[49], M. Colonna[19], A. Comerma-Montells[45], L. Congedo[24], J. Connaughton[57], A. Contu[32], N. Cooke[60], G. Cordova[35,s], C. Coronel[66], I. Corredoira [12], A. Correia[16], G. Corti[49], G. C. Costantino[61], J. Cottee Meldrum[55], B. Couturier[49], D.C. Craik[51], N. Crepet[14], M. Cruz Torres[2,g], M. Cubero Campos[9], E. Curras Rivera[50], R. Currie[59], C.L. Da Silva[68], X. Dai[4], J. Dalseno[44], C. D'Ambrosio[62], G. Darze[3], A. Davidson[57], J.E. Davies[63], O. De Aguiar Francisco[63], C. De Angelis[32], F. De Benedetti[49], J. de Boer[38], K. De Bruyn[83], S. De Capua[63], M. De Cian[63],

U. De Freitas Carneiro Da Graca[2], E. De Lucia[28], J.M. De Miranda[2], L. De Paula[3], M. De Serio[24,h], P. De Simone[28], F. De Vellis[19], J.A. de Vries[84], F. Debernardis[24], D. Decamp[10], S. Dekkers[1], L. Del Buono[16], B. Delaney[65], J. Deng[8], V. Denysenko[51], O. Deschamps[11], F. Dettori[32,k], B. Dey[80], P. Di Nezza[28], S. Ding[69], Y. Ding[50], L. Dittmann[22], A. D. Docheva[60], A. Doheny[57], C. Dong[4], F. Dordei[32], A.C. dos Reis[2], A. D. Dowling[69], L. Dreyfus[13], W. Duan[73], P. Duda[86], L. Dufour[50], V. Duk[34], P. Durante[49], M. M. Duras[86], J.M. Durham[68], O. D. Durmus[80], K. Duwe[49], A. Dziurda[41], S. Easo[58], E. Eckstein[18], U. Egede[1], S. Eisenhardt[59], E. Ejopu[61], L. Eklund[87], M. Elashri[66], D. Elizondo Blanco[9], J. Ellbracht[19], S. Ely[62], A. Ene[43], J. Eschle[69], T. Evans[38], F. Fabiano[14], S. Faghih[66], L.N. Falcao[31,o], B. Fang[7], R. Fantechi[35], L. Fantini[34,r], M. Faria[50], K. Farmer[59], F. Fassin[83,38], D. Fazzini[31,o], L. Felkowski[86], C. Feng[6], M. Feng[5,7], A. Fernandez Casani[48], M. Fernandez Gomez[47], A.D. Fernez[67], F. Ferrari[25,j], F. Ferreira Rodrigues[3], M. Ferrillo[51], M. Ferro-Luzzi[49], R.A. Fini[24], M. Fiorini[26,l], M. Firlej[40], K.L. Fischer[64], D.S. Fitzgerald[89], C. Fitzpatrick[63], T. Fiutowski[40], F. Fleuret[15], A. Fomin[52], M. Fontana[25,49], L. A. Foreman[63], R. Forty[49], D. Foulds-Holt[59], V. Franco Lima[3], M. Franco Sevilla[67], M. Frank[49], E. Franzoso[26,l], G. Frau[63], C. Frei[49], D.A. Friday[63,49], J. Fu[7], Y. Fu[5], Q. Führing[19,f,56], T. Fulghesu[13], G. Galati[24,h], M.D. Galati[38], A. Gallas Torreira[47], D. Galli[25,j], S. Gambetta[59], M. Gandelman[3], P. Gandini[30], B. Ganie[63], H. Gao[7], R. Gao[64], T.Q. Gao[56], Y. Gao[8], Y. Gao[6], Y. Gao[8], L.M. Garcia Martin[50], P. Garcia Moreno[45], J. García Pardiñas[65], P. Gardner[67], L. Garrido[45], C. Gaspar[49], A. Gavrikov[33], E. Gersabeck[20], M. Gersabeck[20], T. Gershon[57], S. Ghizzo[29,m], Z. Ghorbanimoghaddam[55], F. I. Giasemis[16,e], V. Gibson[56], H.K. Giemza[42], A.L. Gilman[66], M. Giovannetti[28], A. Gioventù[47], L. Girardey[63,58], M.A. Giza[41], F.C. Glaser[22,14], V.V. Gligorov[16], C. Göbel[70], L. Golinka-Bezshyyko[88], E. Golobardes[46], A. Golutvin[62,49], S. Gomez Fernandez[45], W. Gomulka[40], F. Goncalves Abrantes[64], I. Gonçales Vaz[49], M. Goncerz[41], G. Gong[4,c], J. A. Gooding[19], C. Gotti[31], E. Govorkova[65], J.P. Grabowski[30], L.A. Granado Cardoso[49], R. Grande Quartieri[2], E. Graugés[45], E. Graverini[35,50], L. Grazette[57], G. Graziani[27], A. T. Grecu[43], N.A. Grieser[66], L. Grillo[60], C. Gu[15], M. Guarise[26], L. Guerry[11], A.-K. Guseinov[50], Y. Guz[6], T. Gys[49], K. Habermann[18], T. Hadavizadeh[1], C. Hadjivasiliou[67], G. Haefeli[50], C. Haen[49], S. Haken[56], G. Hallett[57], P.M. Hamilton[67], Q. Han[33], X. Han[22,49], S. Hansmann-Menzemer[22], N. Harnew[64], T. J. Harris[1], L. Hartman[50], M. Hartmann[14], S. Hashmi[40], J. He[7,d], N. Heatley[14], A. Hedes[63], F. Hemmer[49], C. Henderson[66], R. Henderson[14], R.D.L. Henderson[1], A.M. Hennequin[49], K. Hennessy[61], J. Herd[62], P. Herrero Gascon[22], J. Heuel[17], A. Heyn[13], A. Hicheur[3], G. Hijano Mendizabal[51], J. Horswill[63], R. Hou[8], Y. Hou[11], D.C. Houston[60], N. Howarth[61], W. Hu[7,d], X. Hu[4], W. Hulsbergen[38], R.J. Hunter[57], D. Hurley[57], D. Hutchcroft[61], M. Idzik[40], P. Ilten[66], A. Iohner[10], H. Jage[17], S.J. Jaimes Elles[77,48,49], S. Jakobsen[49], T. Jakoubek[78], E. Jans[38], A. Jawahery[67], C. Jayaweera[54], A. Jelavic[1], V. Jevtic[19], Z. Jia[16], E. Jiang[67], X. Jiang[5,7], Y. Jiang[7], Y. J. Jiang[6], E. Jimenez Moya[9], N. Jindal[91], M. John[64], A. John Rubesh Rajan[23], D. Johnson[54], C.R. Jones[56], S. Joshi[42], B. Jost[49], J. Juan Castella[56], N. Jurik[49], I. Juszczak[41], K. Kalecinska[40], D. Kaminaris[50], S. Kandybei[52], M. Kane[59], Y. Kang[4,c], C. Kar[11], M. Karacson[49], A. Kauniskangas[50], J.W. Kautz[66], M.K. Kazanecki[41], F. Keizer[49], M. Kenzie[56], T. Ketel[38], B. Khanji[69], S. Kholodenko[62,49], G. Khreich[14], F. Kiraz[14], T. Kirn[17], V.S. Kirsebom[31,o], N. Kleijne[35,s], A. Kleimenova[50], D. K. Klekots[88],

K. Klimaszewski[42], M.R. Kmiec[42], T. Knospe[19], R. Kolb[22], S. Koliiev[53], L. Kolk[19], A. Konoplyannikov[6], P. Kopciewicz[49], P. Koppenburg[38], A. Korchin[52], I. Kostiuk[38], O. Kot[53], S. Kotriakhova[32], E. Kowalczyk[67], O. Kravcov[81], M. Kreps[57], W. Krupa[49], W. Krzemien[42], O. Kshyvanskyi[53], S. Kubis[86], M. Kucharczyk[41], A. Kupsc[87,42], V. Kushnir[52], B. Kutsenko[13], J. Kvapil[68], I. Kyryllin[52], D. Lacarrere[49], P. Laguarta Gonzalez[45], A. Lai[32], A. Lampis[32], D. Lancierini[62], C. Landesa Gomez[47], J.J. Lane[1], G. Lanfranchi[28], C. Langenbruch[22], T. Latham[57], F. Lazzari[35,t], C. Lazzeroni[54], R. Le Gac[13], H. Lee[61], R. Lefèvre[11], M. Lehuraux[57], E. Lemos Cid[49], O. Leroy[13], T. Lesiak[41], E. D. Lesser[68], B. Leverington[22], A. Li[4,c], C. Li[4], C. Li[13], H. Li[73], J. Li[8], K. Li[76], L. Li[63], P. Li[7], P.-R. Li[74], Q. Li[5,7], T. Li[72], T. Li[73], W. Li[1], Y. Li[8], Y. Li[5], Y. Li[4], Z. Lian[4,c], Q. Liang[8], X. Liang[69], Z. Liang[32], S. Libralon[48], A. Lightbody[12], T. Lin[58], R. Lindner[49], H. Linton[62], R. Litvinov[66], D. Liu[8], F. L. Liu[1], G. Liu[73], K. Liu[74], S. Liu[5], W. Liu[8], Y. Liu[59], Y. Liu[74], Y. L. Liu[62], G. Loachamin Ordonez[70], I. Lobo[1], A. Lobo Salvia[10], A. Loi[32], T. Long[56], F. C. L. Lopes[2,a], J.H. Lopes[3], A. Lopez Huertas[45], C. Lopez Iribarnegaray[47], Q. Lu[15], C. Lucarelli[49], D. Lucchesi[33,q], M. Lucio Martinez[48], Y. Luo[6], A. Lupato[33,i], M. Lupberger[20], E. Luppi[26,l], K. Lynch[23], S. Lyu[6], X.-R. Lyu[7], H. Ma[72], S. Maccolini[49], F. Machefert[14], F. Maciuc[43], B. Mack[69], I. Mackay[64], L. M. Mackey[69], L.R. Madhan Mohan[56], M. J. Madurai[54], D. Magdalinski[38], J.J. Malczewski[41], S. Malde[64], L. Malentacca[49], G. Manca[32,k], G. Mancinelli[13], C. Mancuso[14], R. Manera Escalero[45], A. Mangalasseri[80], F. M. Manganella[37], D. Manuzzi[25], S. Mao[7], D. Marangotto[30,n], J.F. Marchand[10], R. Marchevski[50], U. Marconi[25], E. Mariani[16], S. Mariani[49], C. Marin Benito[45], J. Marks[22], A.M. Marshall[55], L. Martel[64], G. Martelli[19], G. Martellotti[36], L. Martinazzoli[49], M. Martinelli[31,o], C. Martinez[3], D. Martinez Gomez[83], D. Martinez Santos[44], F. Martinez Vidal[48], A. Martorell i Granollers[46], A. Massafferri[2], R. Matev[49], A. Mathad[49], C. Matteuzzi[69], K.R. Mattioli[15], A. Mauri[62], E. Maurice[15], J. Mauricio[45], P. Mayencourt[50], J. Mazorra de Cos[48], M. Mazurek[42], D. Mazzanti Tarancon[45], M. McCann[62], N.T. McHugh[60], A. McNab[63], R. McNulty[23], B. Meadows[66], D. Melnychuk[42], D. Mendoza Granada[16], P. Menendez Valdes Perez[47], F. M. Meng[4,c], M. Merk[38,84], A. Merli[50,30], L. Meyer Garcia[67], D. Miao[5,7], H. Miao[30], M. Mikhasenko[79], D.A. Milanes[85], A. Minotti[31,o], E. Minucci[28], B. Mitreska[63], D.S. Mitzel[19], R. Mocanu[43], A. Modak[58], L. Moeser[19], R.D. Moise[17], E. F. Molina Cardenas[89], T. Mombächer[47], M. Monk[56], T. Monnard[50], S. Monteil[11], A. Morcillo Gomez[47], G. Morello[28], M.J. Morello[35,s], M.P. Morgenthaler[22], A. Moro[31,o], J. Moron[40], W. Morren[38], A.B. Morris[81,49], A.G. Morris[13], R. Mountain[69], Z. Mu[6], N. Muangkod[65], E. Muhammad[57], F. Muheim[59], M. Mulder[19], K. Müller[51], F. Muñoz-Rojas[9], R. Murta[62], V. Mytrochenko[52], P. Naik[61], T. Nakada[50], R. Nandakumar[58], G. Napoletano[50], I. Nasteva[3], M. Needham[59], N. Neri[30,n], S. Neubert[18], N. Neufeld[49], J. Nicolini[49], D. Nicotra[84], E.M. Niel[15], L. Nisi[19], Q. Niu[74], B. K. Njoki[49], P. Nogarolli[3], P. Nogga[18], C. Normand[47], J. Novoa Fernandez[47], G. Nowak[66], H. N. Nur[60], A. Oblakowska-Mucha[40], T. Oeser[17], O. Okhrimenko[53], R. Oldeman[32,k], F. Oliva[59,49], E. Olivart Pino[45], M. Olocco[19], R.H. O'Neil[49], J.S. Ordonez Soto[11], D. Osthues[19], J.M. Otalora Goicochea[3], P. Owen[51], A. Oyanguren[48], O. Ozcelik[49], F. Paciolla[35,u], A. Padee[42], K.O. Padeken[18], B. Pagare[47], T. Pajero[49], A. Palano[24], L. Palini[30], M. Palutan[28], C. Pan[75], X. Pan[4,c], S. Panebianco[12], S. Paniskaki[49], L. Paolucci[63], A. Papanestis[58], M. Pappagallo[24,h], L.L. Pappalardo[26],

C. Pappenheimer[66], C. Parkes[63], D. Parmar[79], G. Passaleva[27], D. Passaro[35,s], A. Pastore[24], M. Patel[62], J. Patoc[64], C. Patrignani[25,j], A. Paul[69], C.J. Pawley[84], A. Pellegrino[38], J. Peng[5,7], X. Peng[74], M. Pepe Altarelli[28], S. Perazzini[25], H. Pereira Da Costa[68], M. Pereira Martinez[47], A. Pereiro Castro[47], C. Perez[46], P. Perret[11], A. Perrevoort[83], A. Perro[49], M.J. Peters[66], K. Petridis[55], A. Petrolini[29,m], S. Pezzulo[29,m], J. P. Pfaller[66], H. Pham[69], L. Pica[35,s], M. Piccini[34], L. Piccolo[32], B. Pietrzyk[10], R. N. Pilato[61], D. Pinci[36], F. Pisani[49], M. Pizzichemi[31,o,49], V. M. Placinta[43], M. Plo Casasus[47], T. Poeschl[49], F. Polci[16], M. Poli Lener[28], A. Poluektov[13], I. Polyakov[63], E. Polycarpo[3], S. Ponce[49], D. Popov[7,49], K. Popp[19], K. Prasanth[59], C. Prouve[44], D. Provenzano[32,k,49], V. Pugatch[53], A. Puicercus Gomez[49], G. Punzi[35,t], J.R. Pybus[68], Q. Qian[6], W. Qian[7], N. Qin[4,c], R. Quagliani[49], R.I. Rabadan Trejo[57], B. Rachwal[40], R. Racz[81], J.H. Rademacker[55], M. Rama[35], M. Ramírez García[89], V. Ramos De Oliveira[70], M. Ramos Pernas[49], M.S. Rangel[3], G. Raven[39], M. Rebollo De Miguel[48], F. Redi[30,i], J. Reich[55], F. Reiss[20], Z. Ren[7], P.K. Resmi[64], M. Ribalda Galvez[45], R. Ribatti[50], G. Ricart[12], D. Riccardi[35,s], S. Ricciardi[58], K. Richardson[65], M. Richardson-Slipper[56], F. Riehn[19], K. Rinnert[61], P. Robbe[14,49], G. Robertson[60], E. Rodrigues[61], A. Rodriguez Alvarez[45], E. Rodriguez Fernandez[47], J.A. Rodriguez Lopez[77], E. Rodriguez Rodriguez[49], J. Roensch[19], A. Rogovskiy[58], D.L. Rolf[19], P. Roloff[49], V. Romanovskiy[66], A. Romero Vidal[47], G. Romolini[26,49], F. Ronchetti[50], T. Rong[6], M. Rotondo[28], M.S. Rudolph[69], M. Ruiz Diaz[22], J. Ruiz Vidal[84], J. J. Saavedra-Arias[9], J.J. Saborido Silva[47], S. E. R. Sacha Emile R.[49], D. Sahoo[80], N. Sahoo[54], B. Saitta[32], M. Salomoni[31,49,o], I. Sanderswood[48], R. Santacesaria[36], C. Santamarina Rios[47], M. Santimaria[28], L. Santoro [2], E. Santovetti[37], A. Saputi[26,49], A. Sarnatskiy[83], G. Sarpis[49], M. Sarpis[81], C. Satriano[36], A. Satta[37], M. Saur[74], H. Sazak[17], F. Sborzacchi[49,28], A. Scarabotto[19], S. Schael[17], S. Scherl[61], M. Schiller[22], H. Schindler[49], M. Schmelling[21], B. Schmidt[49], N. Schmidt[68], S. Schmitt[65], H. Schmitz[18], O. Schneider[50], A. Schopper[62], N. Schulte[19], M.H. Schune[14], G. Schwering[17], B. Sciascia[28], A. Sciuccati[49], G. Scriven[84], I. Segal[79], S. Sellam[47], M. Senghi Soares[39], A. Sergi[29,m], N. Serra[51], L. Sestini[27], B. Sevilla Sanjuan[46], Y. Shang[6], D.M. Shangase[89], R. S. Sharma[69], L. Shchutska[50], T. Shears[61], J. Shen[6], Z. Shen[38], S. Sheng[50], B. Shi[7], J. Shi[56], Q. Shi[7], W. S. Shi[73], E. Shmanin[25], R. Silva Coutinho[2], G. Simi[33,q], S. Simone[24,h], M. Singha[80], I. Siral[50], N. Skidmore[57], T. Skwarnicki[69], M.W. Slater[54], E. Smith[65], M. Smith[62], L. Soares Lavra[59], M.D. Sokoloff[66], F.J.P. Soler[60], A. Solomin[55], K. Solovieva[20], N. S. Sommerfeld[18], R. Song[1], Y. Song[50], Y. Song[4,c], Y. S. Song[6], F.L. Souza De Almeida[45], B. Souza De Paula[3], K.M. Sowa[40], E. Spadaro Norella[29,m], E. Spedicato[25], J.G. Speer[19], P. Spradlin[60], F. Stagni[49], M. Stahl[79], S. Stahl[49], S. Stanislaus[64], M. Stefaniak[91], O. Steinkamp[51], F. Suljik[64], J. Sun[63], L. Sun[75], M. Sun[6], D. Sundfeld[2], W. Sutcliffe[51], P. Svihra[78], V. Svintozelskyi[48], K. Swientek[40], F. Swystun[56], A. Szabelski[42], T. Szumlak[40], Y. Tan[7], Y. Tang[75], Y. T. Tang[7], M.D. Tat[22], J. A. Teijeiro Jimenez[47], F. Terzuoli[35,u], F. Teubert[49], E. Thomas[49], D.J.D. Thompson[54], A. R. Thomson-Strong[59], H. Tilquin[62], V. Tisserand[11], S. T'Jampens[10], M. Tobin[5,49], T. T. Todorov[20], L. Tomassetti[26,l], G. Tonani[30], X. Tong[6], T. Tork[30], L. Toscano[19], D.Y. Tou[4,c], C. Trippl[46], G. Tuci[22], N. Tuning[38], L.H. Uecker[22], A. Ukleja[40], A. Upadhyay[49], B. Urbach[59], A. Usachov[38], U. Uwer[22], V. Vagnoni[25,49], A. Vaitkevicius[81], V. Valcarce Cadenas[47], G. Valenti[25], N. Valls Canudas[49], J. van Eldik[49],

H. Van Hecke[68], E. van Herwijnen[62], C.B. Van Hulse[47,w], R. Van Laak[50], M. van Veghel[84], G. Vasquez[51], R. Vazquez Gomez[45], P. Vazquez Regueiro[47], C. Vázquez Sierra[44], S. Vecchi[26], J. Velilla Serna[48], J.J. Velthuis[55], M. Veltri[27,v], A. Venkateswaran[50], M. Verdoglia[32], M. Vesterinen[57], W. Vetens[69], D. Vico Benet[64], P. Vidrier Villalba[45], M. Vieites Diaz[47], X. Vilasis-Cardona[46], E. Vilella Figueras[61], A. Villa[50], P. Vincent[16], B. Vivacqua[3], F.C. Volle[54], D. vom Bruch[13], K. Vos[84], C. Vrahas[59], J. Wagner[19], J. Walsh[35], N. Walter[49], E.J. Walton[1], G. Wan[6], A. Wang[7], B. Wang[5], C. Wang[22], G. Wang[8], H. Wang[74], J. Wang[7], J. Wang[5], J. Wang[4,c], J. Wang[75], M. Wang[49], N. W. Wang[7], R. Wang[55], X. Wang[8], X. Wang[73], X. Wang[4], X. W. Wang[62], Y. Wang[76], Y. Wang[6], Y. H. Wang[74], Z. Wang[14], Z. Wang[30], J.A. Ward[57,1], M. Waterlaat[49], N.K. Watson[54], D. Websdale[62], Y. Wei[6], Z. Weida[7], J. Wendel[44], B.D.C. Westhenry[55], C. White[56], M. Whitehead[60], E. Whiter[54], A.R. Wiederhold[63], D. Wiedner[19], M. A. Wiegertjes[38], C. Wild[64], G. Wilkinson[64], M.K. Wilkinson[66], M. Williams[65], M. J. Williams[49], M.R.J. Williams[59], R. Williams[56], S. Williams[55], Z. Williams[55], F.F. Wilson[58], M. Winn[12], W. Wislicki[42], M. Witek[41], L. Witola[19], T. Wolf[22], E. Wood[56], G. Wormser[14], S.A. Wotton[56], H. Wu[69], J. Wu[8], X. Wu[75], Y. Wu[6,56], Z. Wu[7], K. Wyllie[49], S. Xian[73], Z. Xiang[5], Y. Xie[8], T. X. Xing[30], A. Xu[35,s], L. Xu[4,c], M. Xu[49], R. Xu[89], Z. Xu[49], Z. Xu[92], Z. Xu[7], Z. Xu[5], S. Yadav[26], K. Yang[62], X. Yang[6], Y. Yang[7], Y. Yang[80], Z. Yang[6], Z. Yang[4], H. Yeung[63], H. Yin[8], X. Yin[7], C. Y. Yu[6], J. Yu[72], X. Yuan[5], Y Yuan[5,7], J. A. Zamora Saa[71], M. Zavertyaev[21], M. Zdybal[41], F. Zenesini[25], C. Zeng[5,7], M. Zeng[4,c], S.H Zeng[55], C. Zhang[6], C. Zhang[61], D. Zhang[8], J. Zhang[42], L. Zhang[4,c], R. Zhang[8], S. Zhang[64], S. L. Zhang[72], Y. Zhang[6], Z. Zhang[4,c], J. Zhao[7], Y. Zhao[22], A. Zhelezov[22], S. Z. Zheng[6], X. Z. Zheng[4,c], Y. Zheng[7], T. Zhou[41], X. Zhou[8], V. Zhovkovska[57], L. Z. Zhu[59], X. Zhu[4,c], X. Zhu[8], Y. Zhu[17], V. Zhukov[17], J. Zhuo[48], D. Zuliani[33,q], G. Zunica[28].

[1]*School of Physics and Astronomy, Monash University, Melbourne, Australia*
[2]*Centro Brasileiro de Pesquisas Físicas (CBPF), Rio de Janeiro, Brazil*
[3]*Universidade Federal do Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil*
[4]*Department of Engineering Physics, Tsinghua University, Beijing, China*
[5]*Institute Of High Energy Physics (IHEP), Beijing, China*
[6]*School of Physics State Key Laboratory of Nuclear Physics and Technology, Peking University, Beijing, China*
[7]*University of Chinese Academy of Sciences, Beijing, China*
[8]*Institute of Particle Physics, Central China Normal University, Wuhan, Hubei, China*
[9]*Consejo Nacional de Rectores (CONARE), San Jose, Costa Rica*
[10]*Université Savoie Mont Blanc, CNRS, IN2P3-LAPP, Annecy, France*
[11]*Université Clermont Auvergne, CNRS/IN2P3, LPC, Clermont-Ferrand, France*
[12]*Université Paris-Saclay, Centre d'Etudes de Saclay (CEA), IRFU, Gif-Sur-Yvette, France*
[13]*Aix Marseille Univ, CNRS/IN2P3, CPPM, Marseille, France*
[14]*Université Paris-Saclay, CNRS/IN2P3, IJCLab, Orsay, France*
[15]*Laboratoire Leprince-Ringuet, CNRS/IN2P3, Ecole Polytechnique, Institut Polytechnique de Paris, Palaiseau, France*
[16]*Laboratoire de Physique Nucléaire et de Hautes Énergies (LPNHE), Sorbonne Université, CNRS/IN2P3, Paris, France*
[17]*I. Physikalisches Institut, RWTH Aachen University, Aachen, Germany*
[18]*Universität Bonn - Helmholtz-Institut für Strahlen und Kernphysik, Bonn, Germany*
[19]*Fakultät Physik, Technische Universität Dortmund, Dortmund, Germany*
[20]*Physikalisches Institut, Albert-Ludwigs-Universität Freiburg, Freiburg, Germany*
[21]*Max-Planck-Institut für Kernphysik (MPIK), Heidelberg, Germany*

[22] *Physikalisches Institut, Ruprecht-Karls-Universität Heidelberg, Heidelberg, Germany*
[23] *School of Physics, University College Dublin, Dublin, Ireland*
[24] *INFN Sezione di Bari, Bari, Italy*
[25] *INFN Sezione di Bologna, Bologna, Italy*
[26] *INFN Sezione di Ferrara, Ferrara, Italy*
[27] *INFN Sezione di Firenze, Firenze, Italy*
[28] *INFN Laboratori Nazionali di Frascati, Frascati, Italy*
[29] *INFN Sezione di Genova, Genova, Italy*
[30] *INFN Sezione di Milano, Milano, Italy*
[31] *INFN Sezione di Milano-Bicocca, Milano, Italy*
[32] *INFN Sezione di Cagliari, Monserrato, Italy*
[33] *INFN Sezione di Padova, Padova, Italy*
[34] *INFN Sezione di Perugia, Perugia, Italy*
[35] *INFN Sezione di Pisa, Pisa, Italy*
[36] *INFN Sezione di Roma La Sapienza, Roma, Italy*
[37] *INFN Sezione di Roma Tor Vergata, Roma, Italy*
[38] *Nikhef National Institute for Subatomic Physics, Amsterdam, Netherlands*
[39] *Nikhef National Institute for Subatomic Physics and VU University Amsterdam, Amsterdam, Netherlands*
[40] *AGH - University of Krakow, Faculty of Physics and Applied Computer Science, Kraków, Poland*
[41] *Henryk Niewodniczanski Institute of Nuclear Physics Polish Academy of Sciences, Kraków, Poland*
[42] *National Center for Nuclear Research (NCBJ), Warsaw, Poland*
[43] *Horia Hulubei National Institute of Physics and Nuclear Engineering, Bucharest-Magurele, Romania*
[44] *Universidade da Coruña, A Coruña, Spain*
[45] *ICCUB, Universitat de Barcelona, Barcelona, Spain*
[46] *La Salle, Universitat Ramon Llull, Barcelona, Spain*
[47] *Instituto Galego de Física de Altas Enerxías (IGFAE), Universidade de Santiago de Compostela, Santiago de Compostela, Spain*
[48] *Instituto de Fisica Corpuscular, Centro Mixto Universidad de Valencia - CSIC, Valencia, Spain*
[49] *European Organization for Nuclear Research (CERN), Geneva, Switzerland*
[50] *Institute of Physics, Ecole Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland*
[51] *Physik-Institut, Universität Zürich, Zürich, Switzerland*
[52] *NSC Kharkiv Institute of Physics and Technology (NSC KIPT), Kharkiv, Ukraine*
[53] *Institute for Nuclear Research of the National Academy of Sciences (KINR), Kyiv, Ukraine*
[54] *School of Physics and Astronomy, University of Birmingham, Birmingham, United Kingdom*
[55] *H.H. Wills Physics Laboratory, University of Bristol, Bristol, United Kingdom*
[56] *Cavendish Laboratory, University of Cambridge, Cambridge, United Kingdom*
[57] *Department of Physics, University of Warwick, Coventry, United Kingdom*
[58] *STFC Rutherford Appleton Laboratory, Didcot, United Kingdom*
[59] *School of Physics and Astronomy, University of Edinburgh, Edinburgh, United Kingdom*
[60] *School of Physics and Astronomy, University of Glasgow, Glasgow, United Kingdom*
[61] *Oliver Lodge Laboratory, University of Liverpool, Liverpool, United Kingdom*
[62] *Imperial College London, London, United Kingdom*
[63] *Department of Physics and Astronomy, University of Manchester, Manchester, United Kingdom*
[64] *Department of Physics, University of Oxford, Oxford, United Kingdom*
[65] *Massachusetts Institute of Technology, Cambridge, MA, United States*
[66] *University of Cincinnati, Cincinnati, OH, United States*
[67] *University of Maryland, College Park, MD, United States*
[68] *Los Alamos National Laboratory (LANL), Los Alamos, NM, United States*
[69] *Syracuse University, Syracuse, NY, United States*
[70] *Pontifícia Universidade Católica do Rio de Janeiro (PUC-Rio), Rio de Janeiro, Brazil, associated to* [3]
[71] *Universidad Andres Bello, Santiago, Chile, associated to* [51]
[72] *School of Physics and Electronics, Hunan University, Changsha City, China, associated to* [8]
[73] *State Key Laboratory of Nuclear Physics and Technology, South China Normal University, Guangzhou, China, associated to* [4]
[74] *Lanzhou University, Lanzhou, China, associated to* [5]

[75]*School of Physics and Technology, Wuhan University, Wuhan, China, associated to* [4]
[76]*Henan Normal University, Xinxiang, China, associated to* [8]
[77]*Departamento de Fisica , Universidad Nacional de Colombia, Bogota, Colombia, associated to* [16]
[78]*Institute of Physics of the Czech Academy of Sciences, Prague, Czech Republic, associated to* [63]
[79]*Ruhr Universitaet Bochum, Fakultaet f. Physik und Astronomie, Bochum, Germany, associated to* [19]
[80]*Eotvos Lorand University, Budapest, Hungary, associated to* [49]
[81]*Faculty of Physics, Vilnius University, Vilnius, Lithuania, associated to* [20]
[82]*Institute of Physics and Technology, Ulan Bator, Mongolia, associated to* [5]
[83]*Van Swinderen Institute, University of Groningen, Groningen, Netherlands, associated to* [38]
[84]*Universiteit Maastricht, Maastricht, Netherlands, associated to* [38]
[85]*Universidad de Ingeniería y Tecnología (UTEC), Lima, Peru, associated to* [65]
[86]*Tadeusz Kosciuszko Cracow University of Technology, Cracow, Poland, associated to* [41]
[87]*Department of Physics and Astronomy, Uppsala University, Uppsala, Sweden, associated to* [60]
[88]*Taras Schevchenko University of Kyiv, Faculty of Physics, Kyiv, Ukraine, associated to* [14]
[89]*University of Michigan, Ann Arbor, MI, United States, associated to* [69]
[90]*Indiana University, Bloomington, United States, associated to* [68]
[91]*Ohio State University, Columbus, United States, associated to* [68]
[92]*Kent State University Physics Department, Kent, United States, associated to* [68]

[a]*Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil*
[b]*Department of Physics and Astronomy, University of Victoria, Victoria, Canada*
[c]*Center for High Energy Physics, Tsinghua University, Beijing, China*
[d]*Hangzhou Institute for Advanced Study, UCAS, Hangzhou, China*
[e]*LIP6, Sorbonne Université, Paris, France*
[f]*Lamarr Institute for Machine Learning and Artificial Intelligence, Dortmund, Germany*
[g]*Universidad Nacional Autónoma de Honduras, Tegucigalpa, Honduras*
[h]*Università di Bari, Bari, Italy*
[i]*Università di Bergamo, Bergamo, Italy*
[j]*Università di Bologna, Bologna, Italy*
[k]*Università di Cagliari, Cagliari, Italy*
[l]*Università di Ferrara, Ferrara, Italy*
[m]*Università di Genova, Genova, Italy*
[n]*Università degli Studi di Milano, Milano, Italy*
[o]*Università degli Studi di Milano-Bicocca, Milano, Italy*
[p]*Università di Modena e Reggio Emilia, Modena, Italy*
[q]*Università di Padova, Padova, Italy*
[r]*Università di Perugia, Perugia, Italy*
[s]*Scuola Normale Superiore, Pisa, Italy*
[t]*Università di Pisa, Pisa, Italy*
[u]*Università di Siena, Siena, Italy*
[v]*Università di Urbino, Urbino, Italy*
[w]*Universidad de Alcalá, Alcalá de Henares , Spain*

[†]*Deceased*